

MULTIPLE TEST

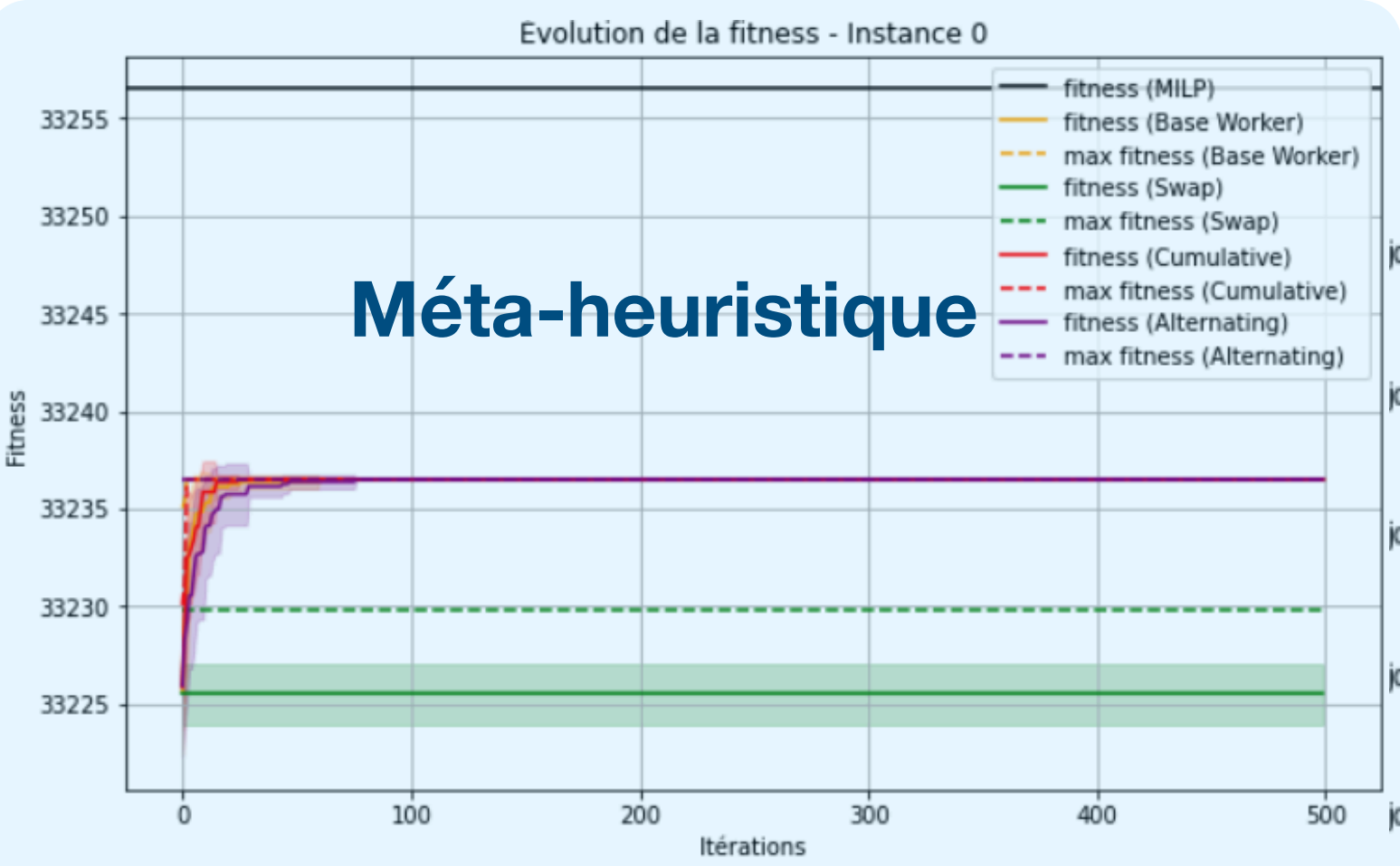
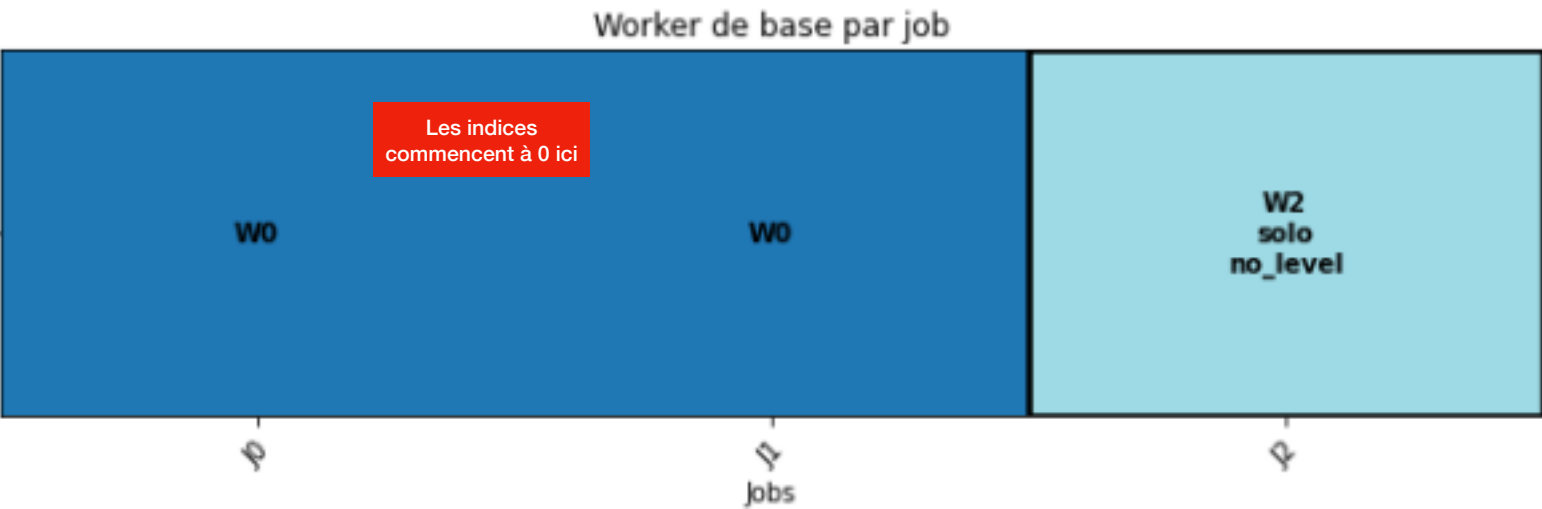
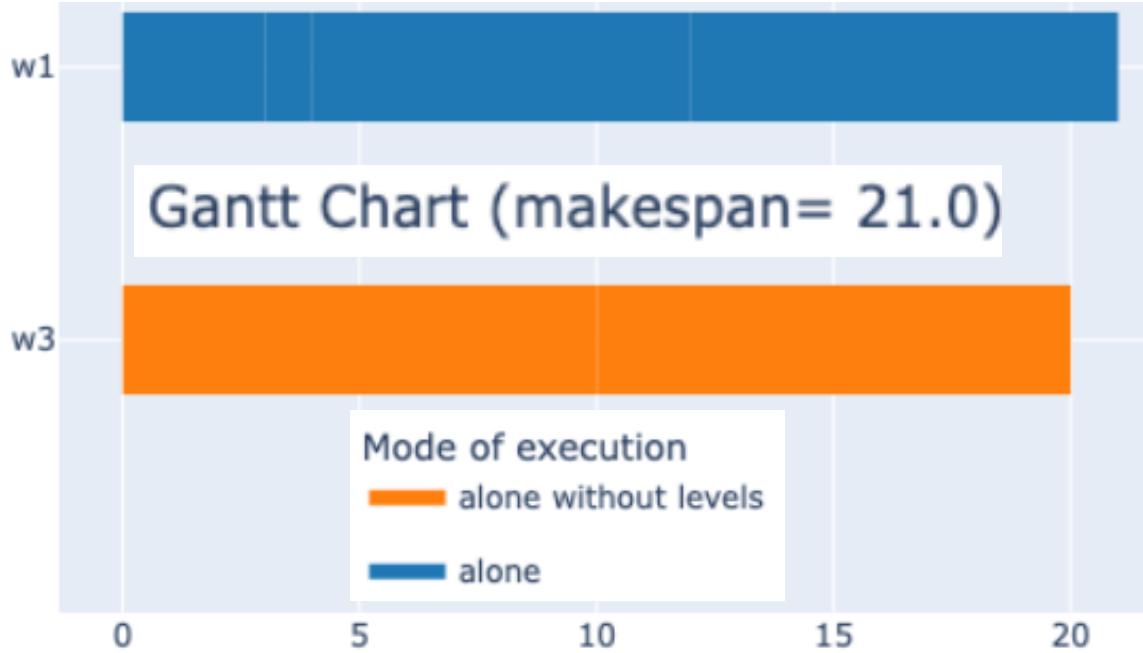
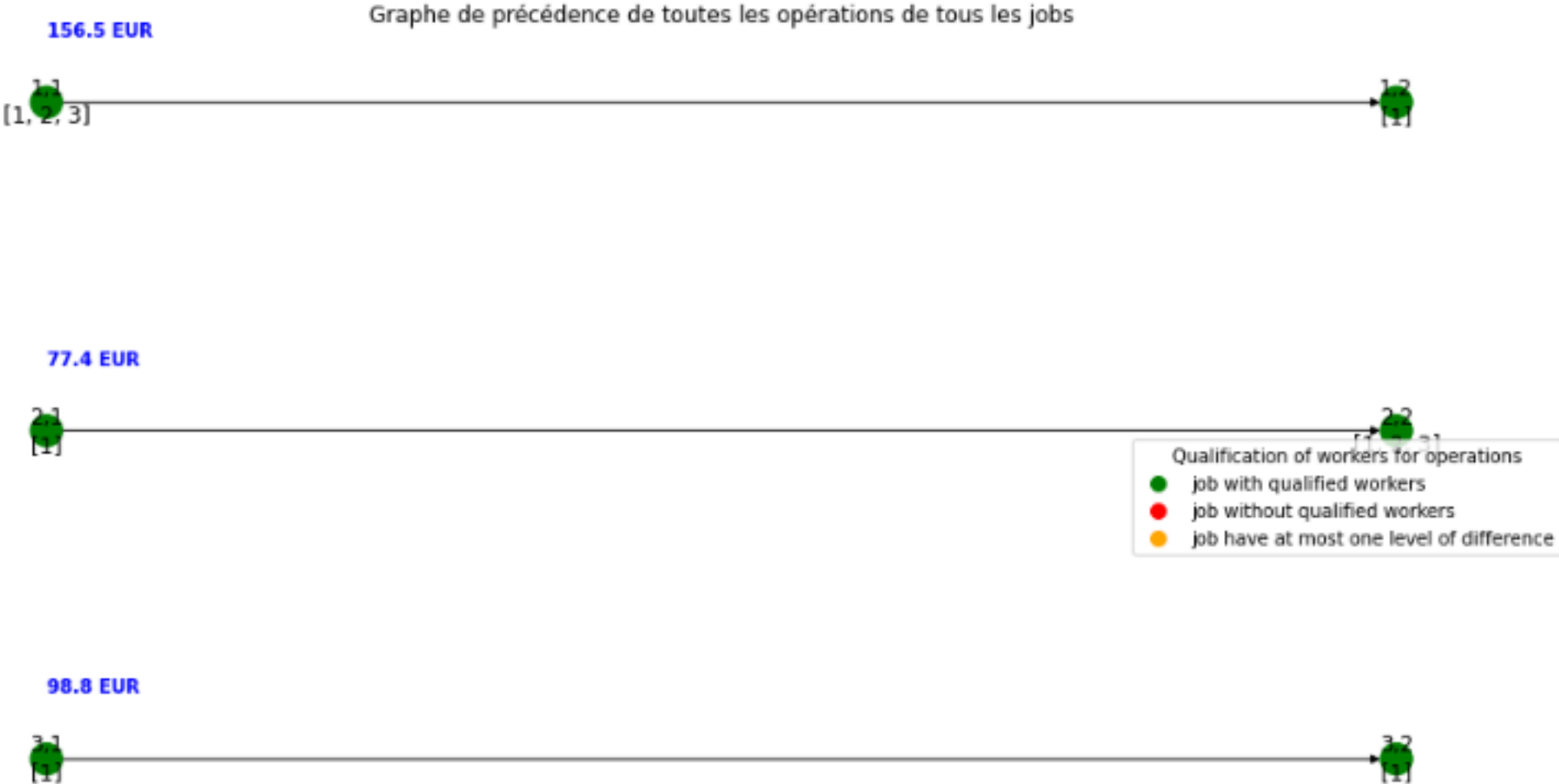
```
CONSTRAINTS_CONFIG = {  
    "job_with_no_skills": True,  
    "constrained_makespan": False,  
    "no_base_worker": False,  
    "no_solo_tasks": False,  
}  
  
WEIGHTS = {"profit": 100, "skills": 10, "cognitive_load": 1}  
  
OBJECTIVE_TERMS_CONFIG = {  
    "benefit_all_jobs": 0,  
    "penalty_levels": 0,  
    "penalty_makespan_job": 0,  
    "Cmax": 1,  
}
```

Note a moi même:
Faire commencer les indices de chaque graphique identiquement

MILP MODEL

instance : 0
profit : 332.67493007272844
skills : 1.0
cognitive_load : 0.0
Cmax : 21.0
33256.493007272846

Instance 0



Comparaison des meilleures valeurs obtenues - Instance 0

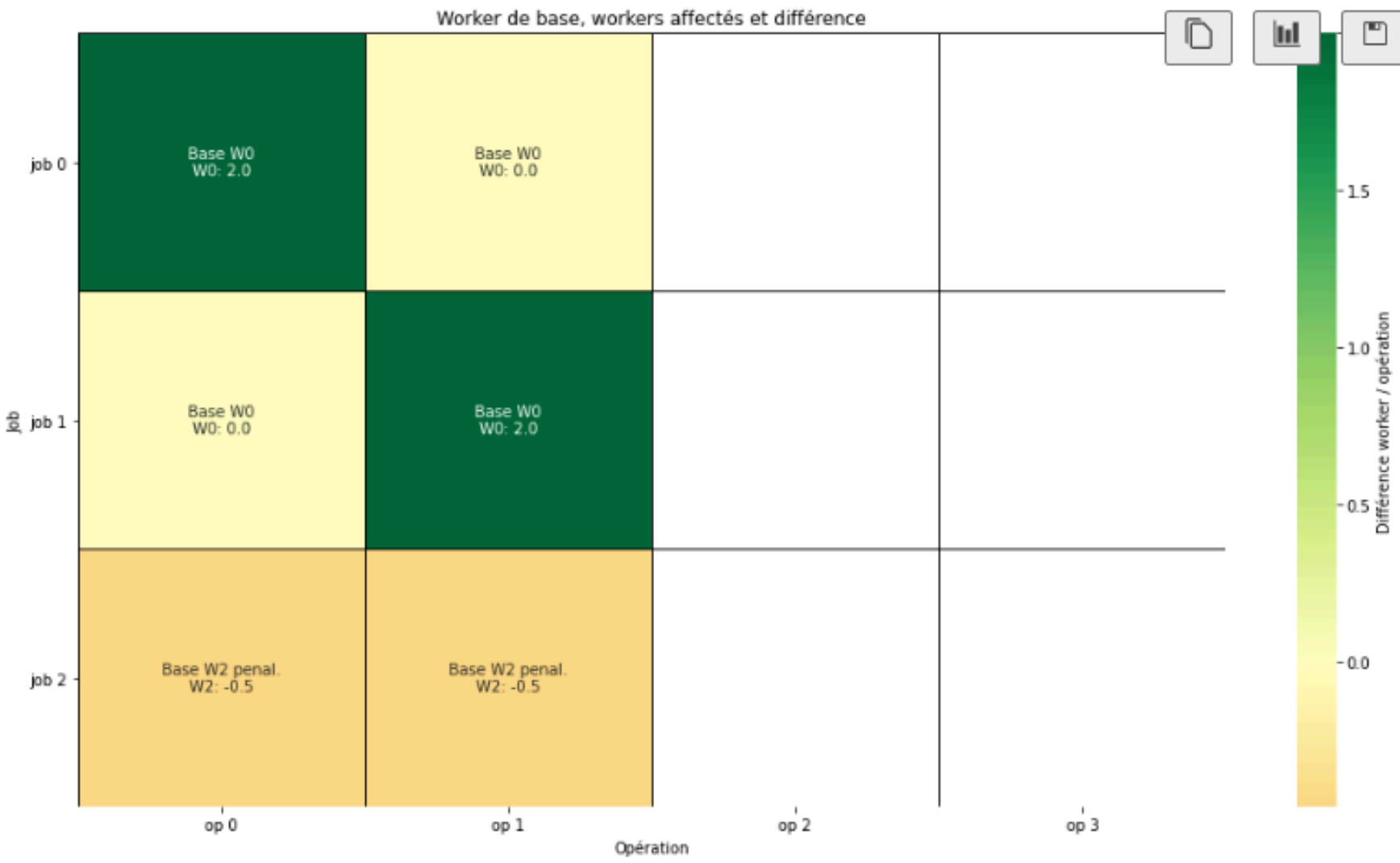
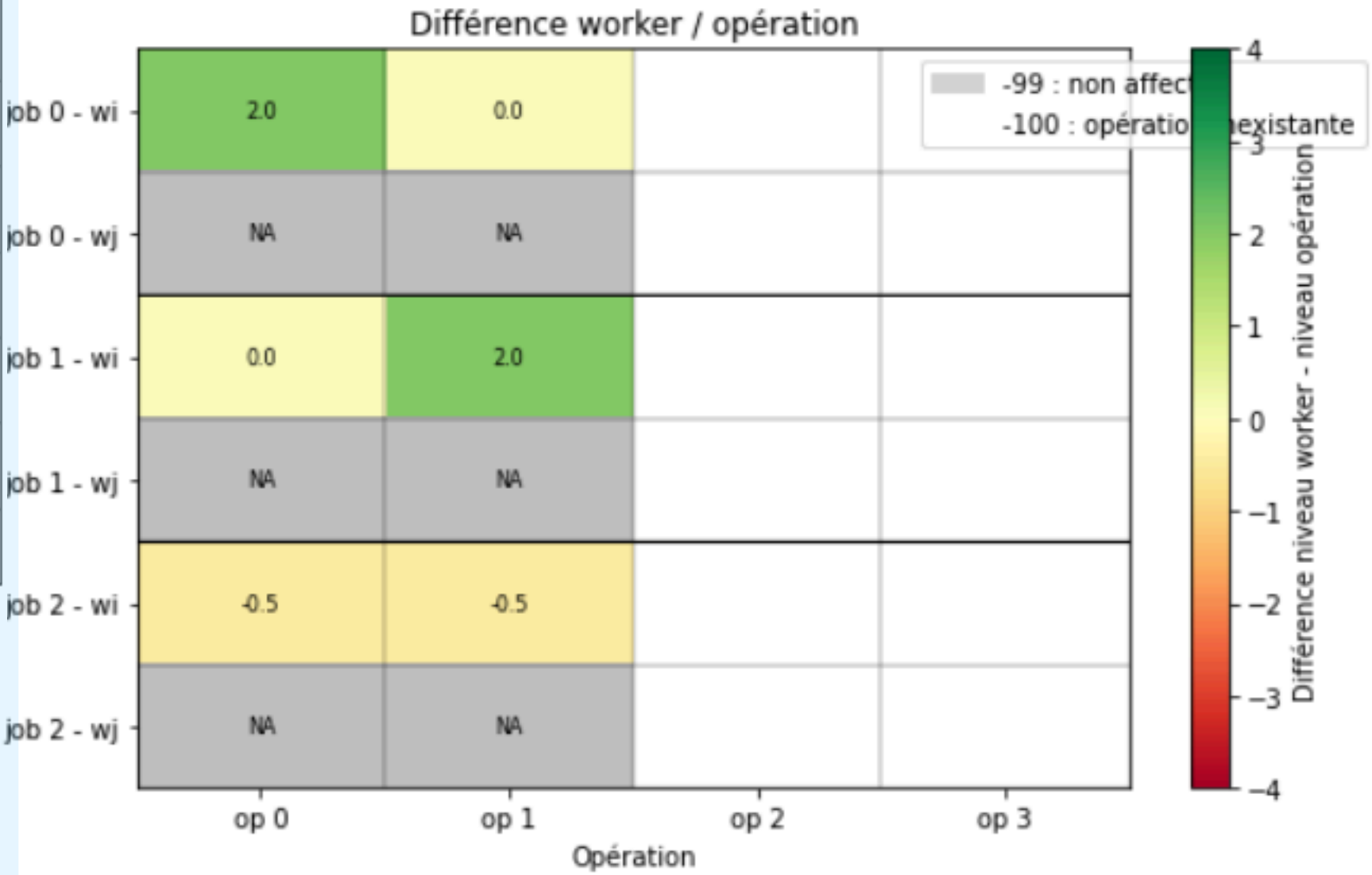
Solution exacte MILP
33256.49

Base worker
Max = 33236.49
Moy. fin = 33236.49
Gap max = 0.06 %
Gap moy fin = 0.06 %

Swap
Max = 33229.76
Moy. fin = 33225.49
Gap max = 0.08 %
Gap moy fin = 0.09 %

Cumulative
Max = 33236.49
Moy. fin = 33236.49
Gap max = 0.06 %
Gap moy fin = 0.06 %

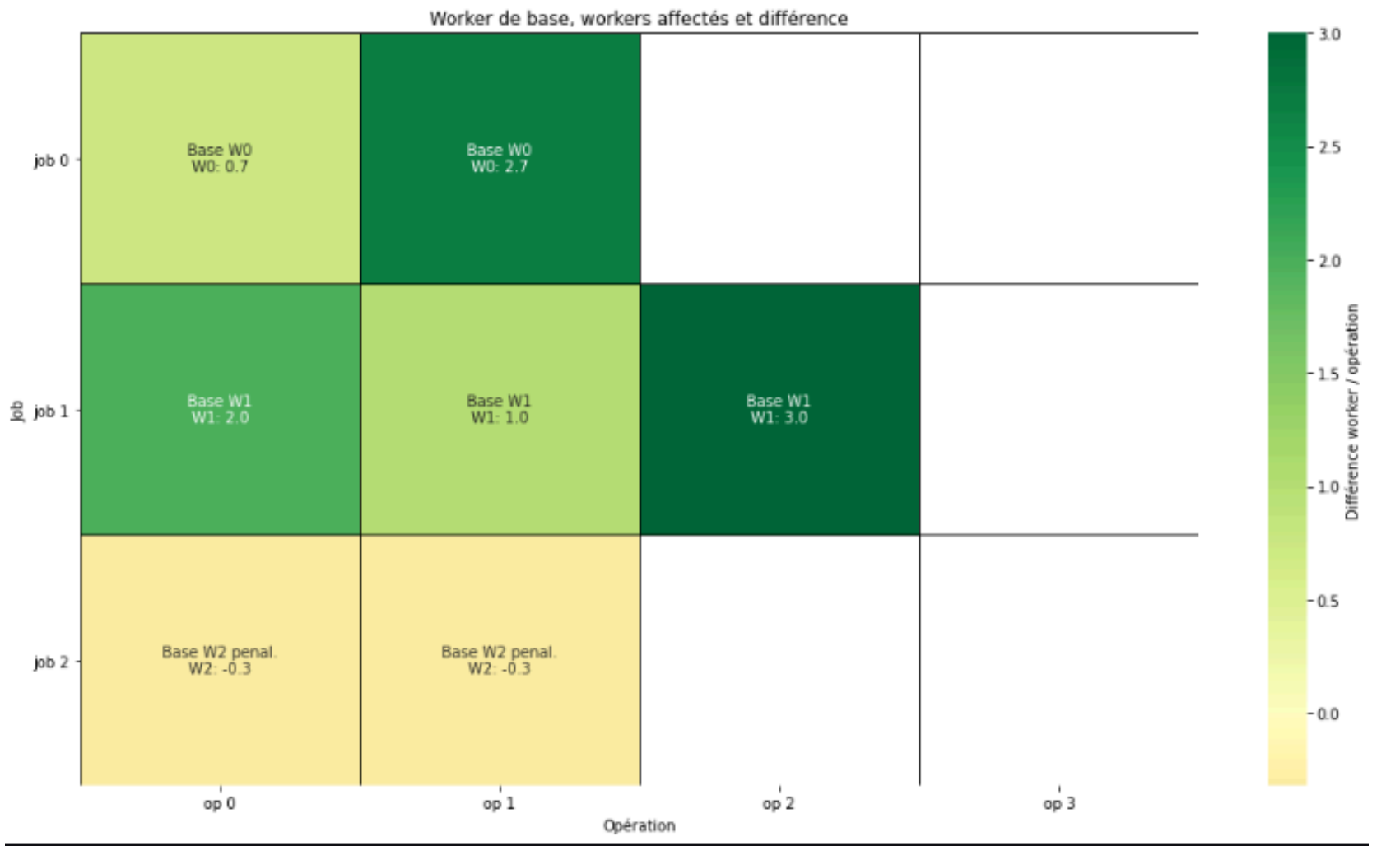
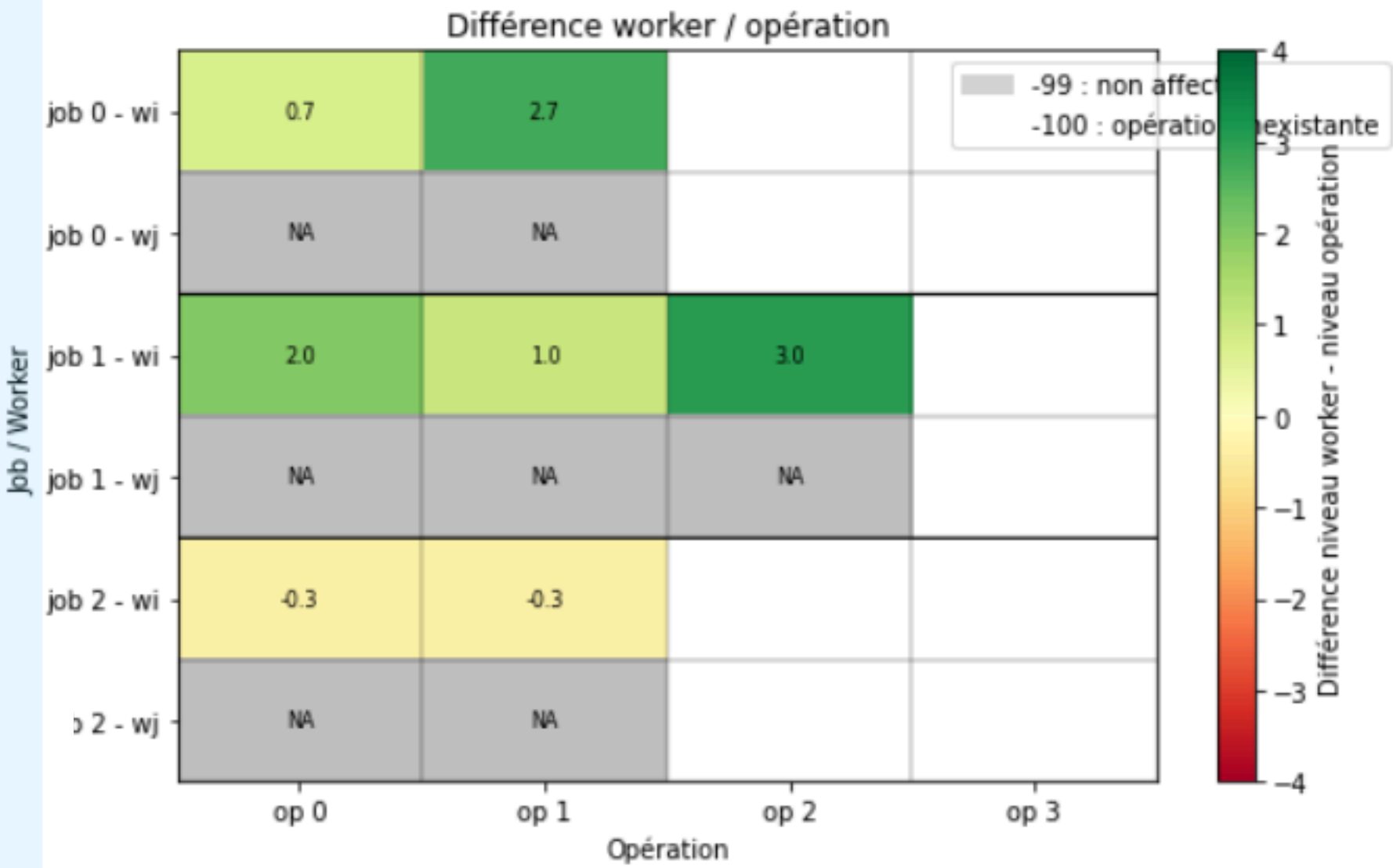
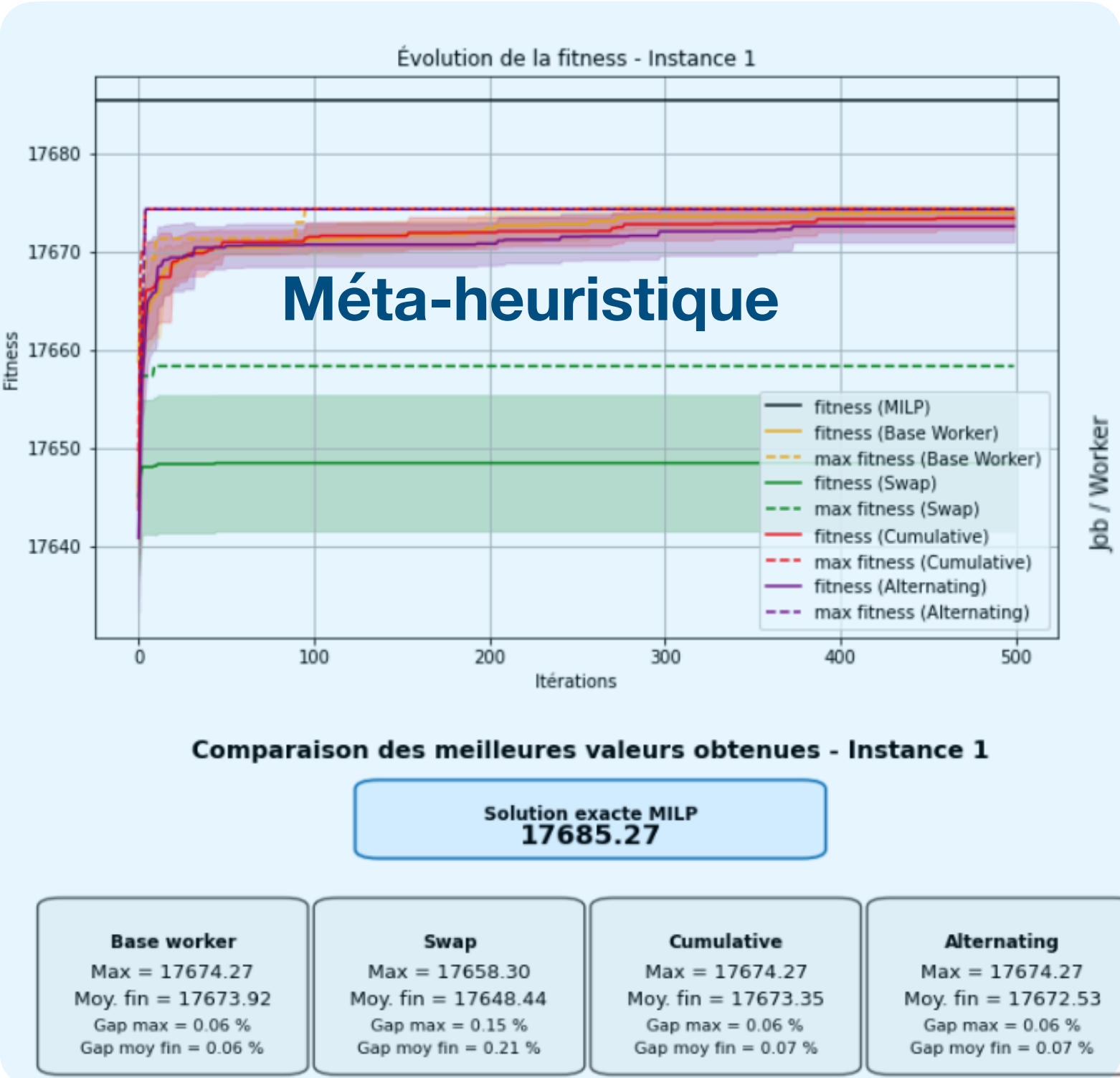
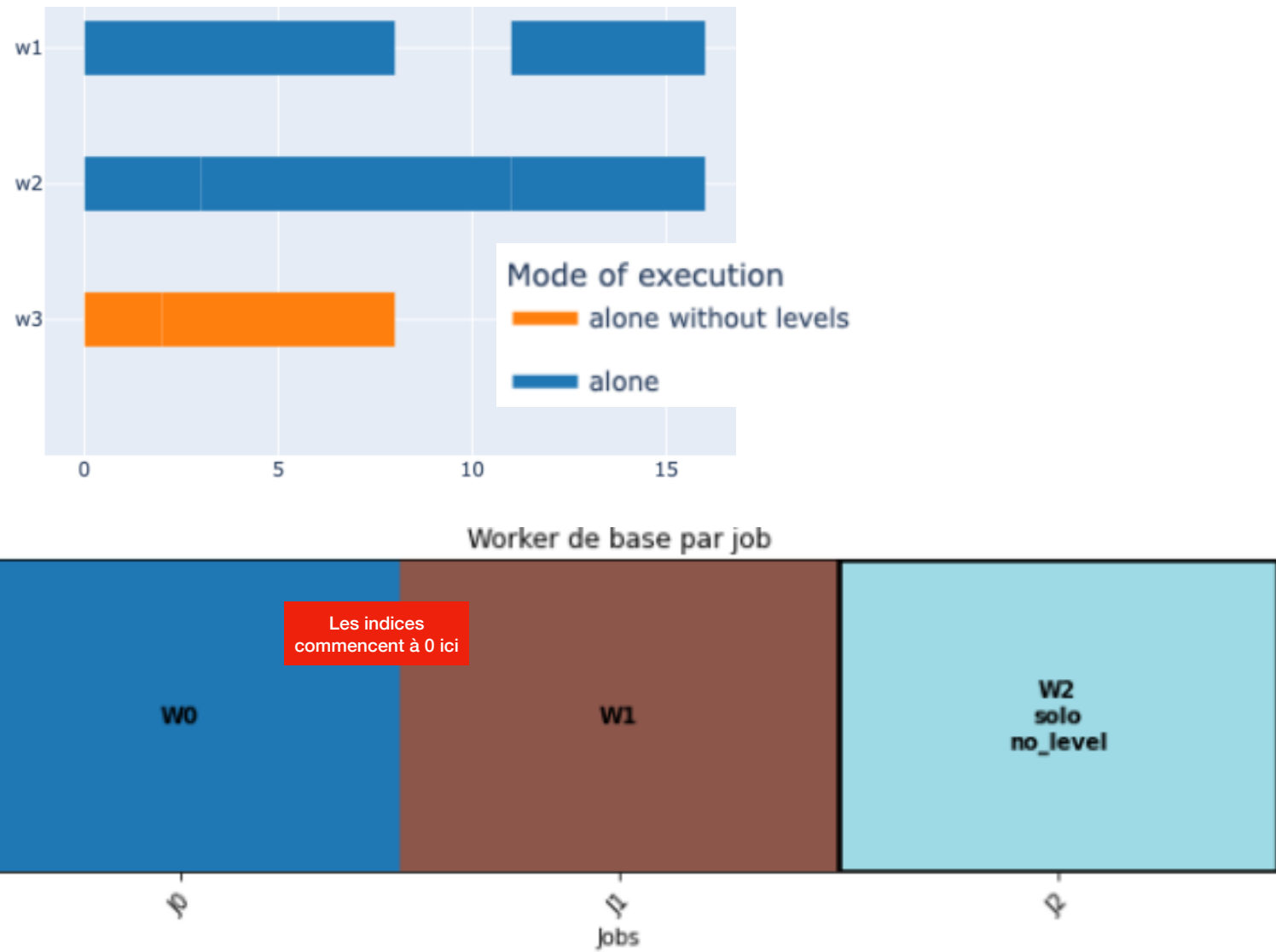
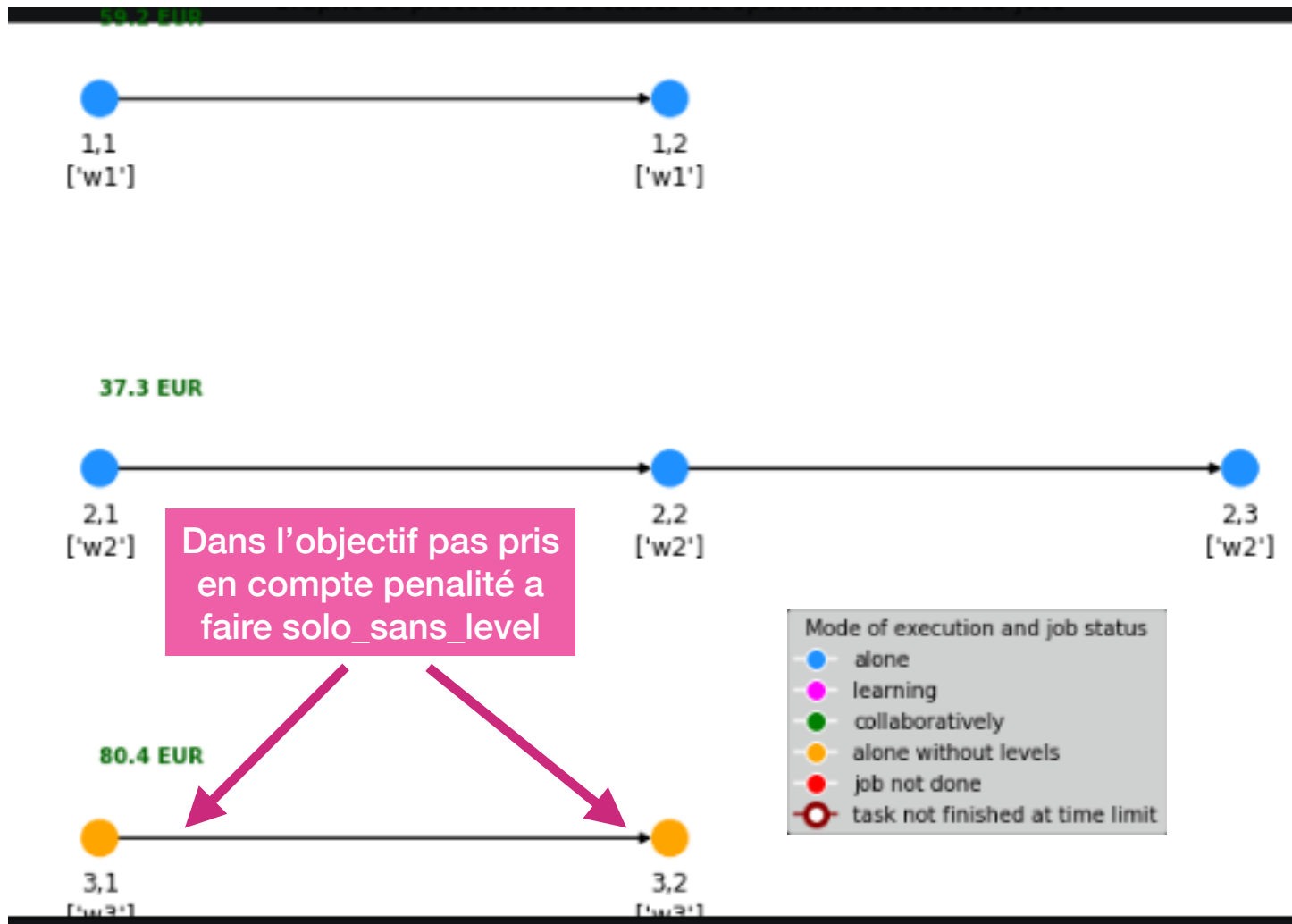
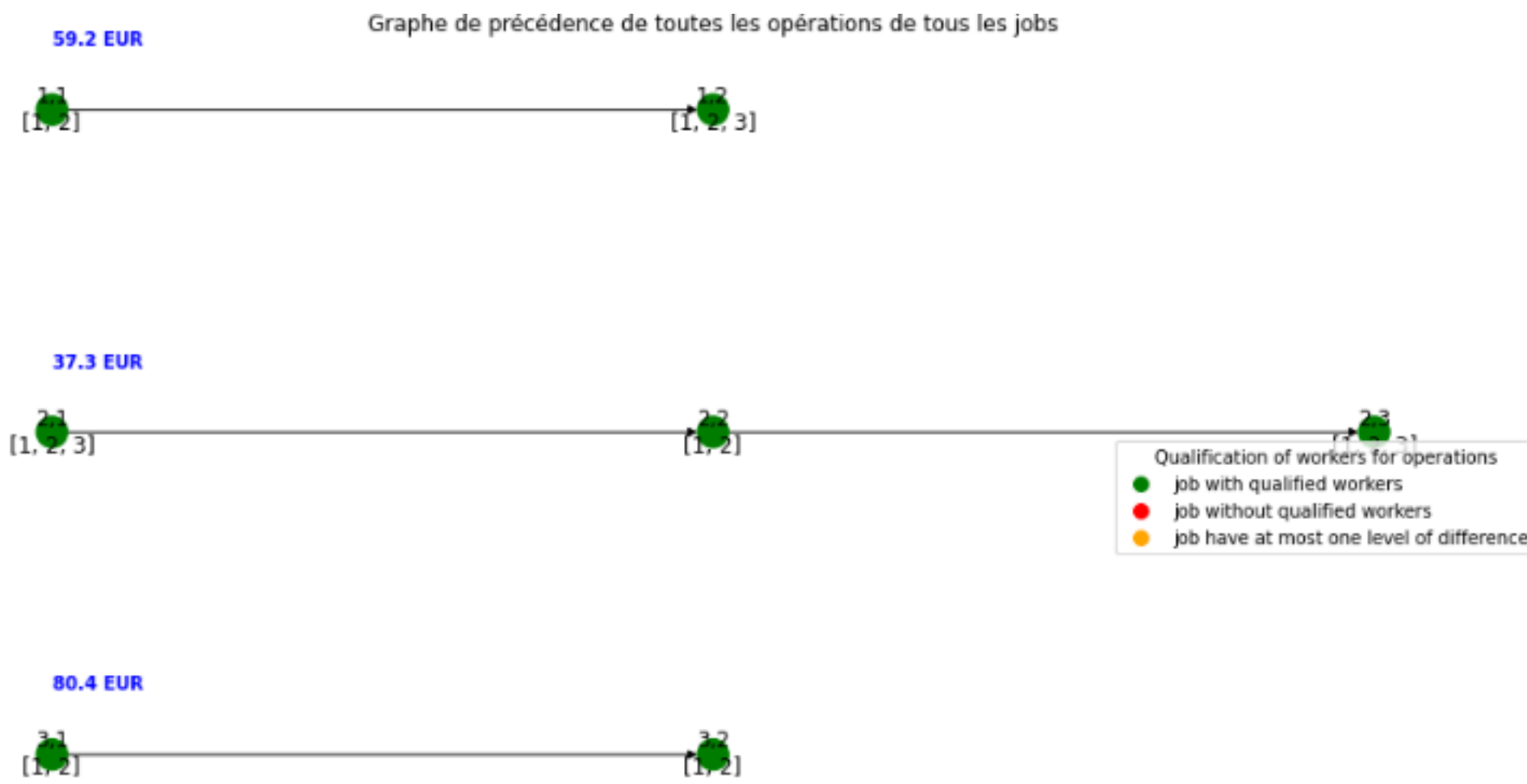
Alternating
Max = 33236.49
Moy. fin = 33236.49
Gap max = 0.06 %
Gap moy fin = 0.06 %



MILP MODEL

instance : 1
profit : 176.91265678978039
skills : 1.0
cognitive_load : 0.0
Cmax : 16.0
17685.265678978038

Instance 1



Comparaison des meilleures valeurs obtenues - Instance 1

Instance 1

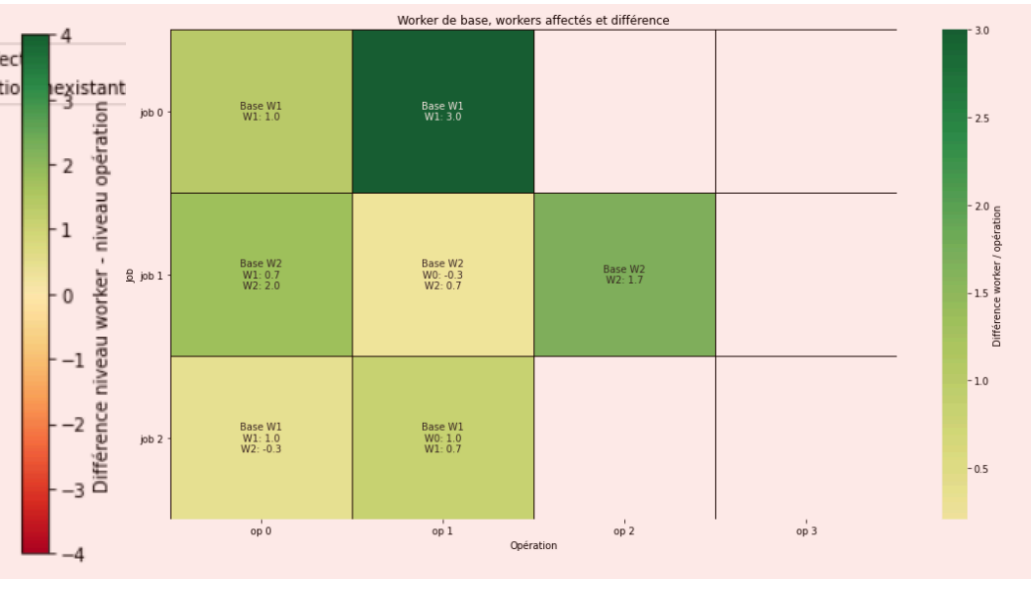
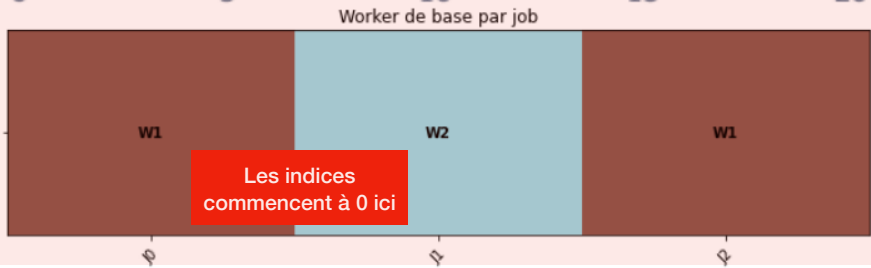
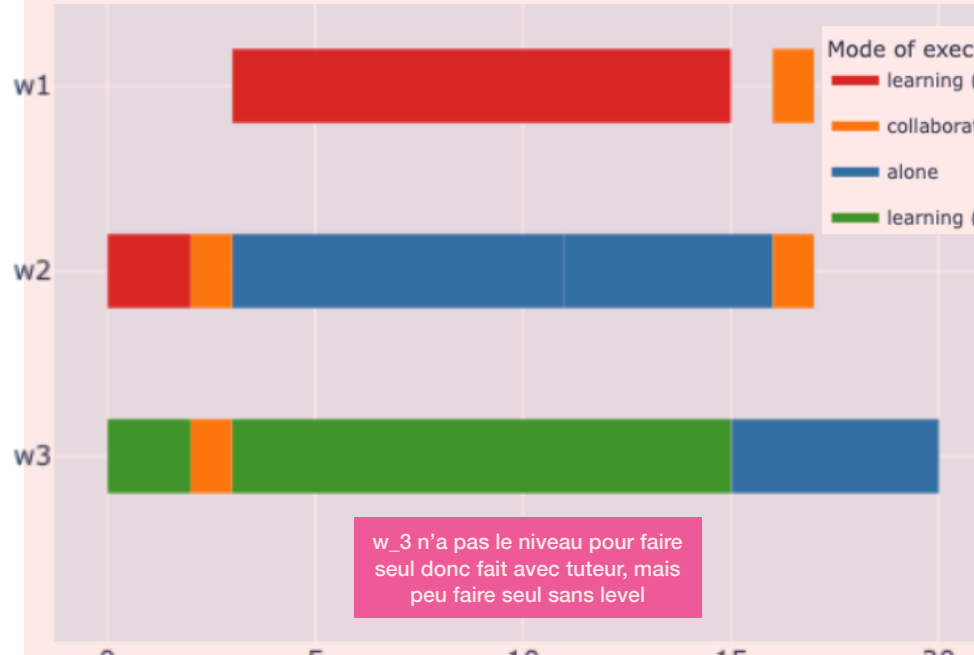
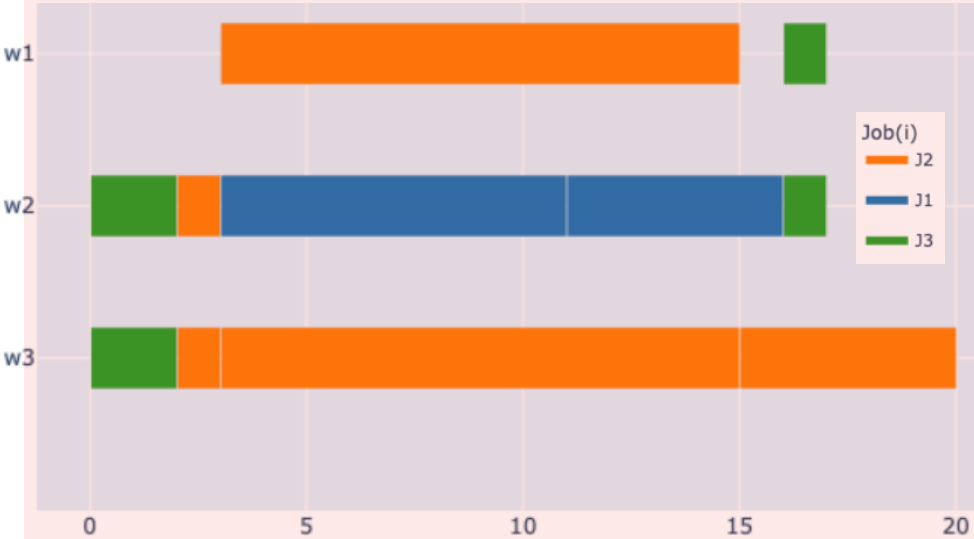
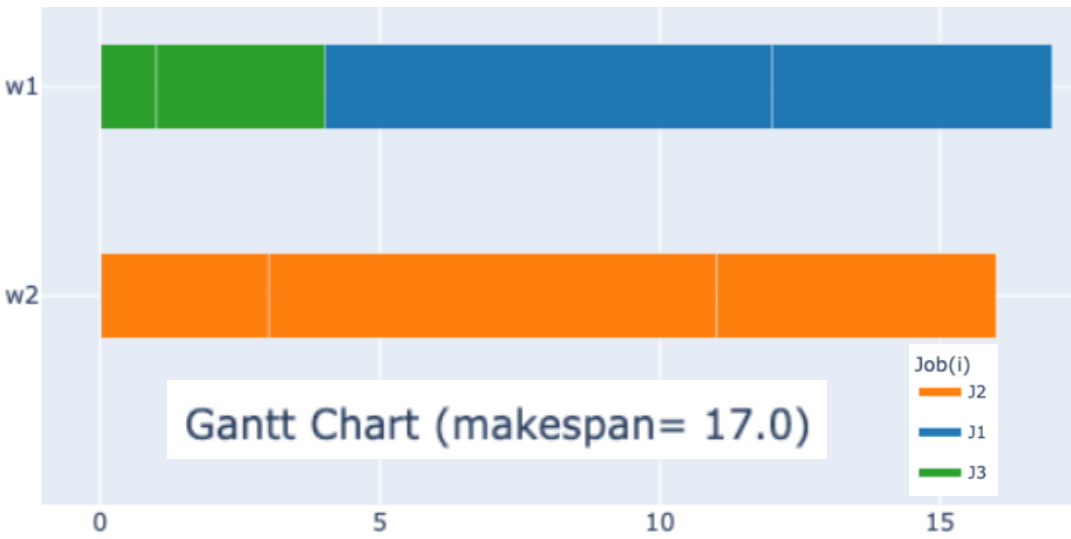
Solution exacte MILP
17685.27

Base worker
Max = 17674.27
Moy. fin = 17673.92
Gap max = 0.06 %
Gap moy fin = 0.06 %

Swap
Max = 17658.30
Moy. fin = 17648.44
Gap max = 0.15 %
Gap moy fin = 0.21 %

Cumulative
Max = 17674.27
Moy. fin = 17673.35
Gap max = 0.06 %
Gap moy fin = 0.07 %

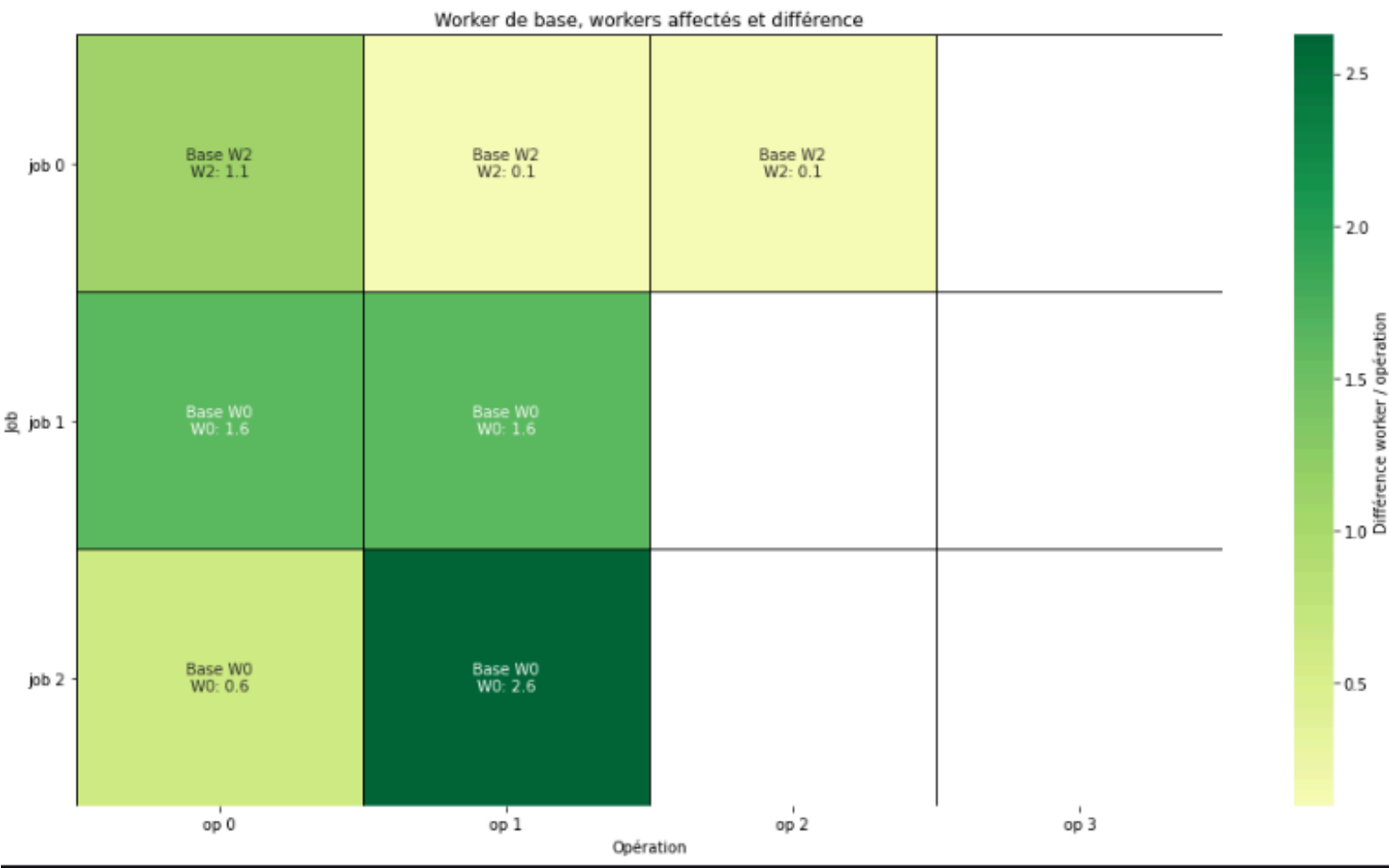
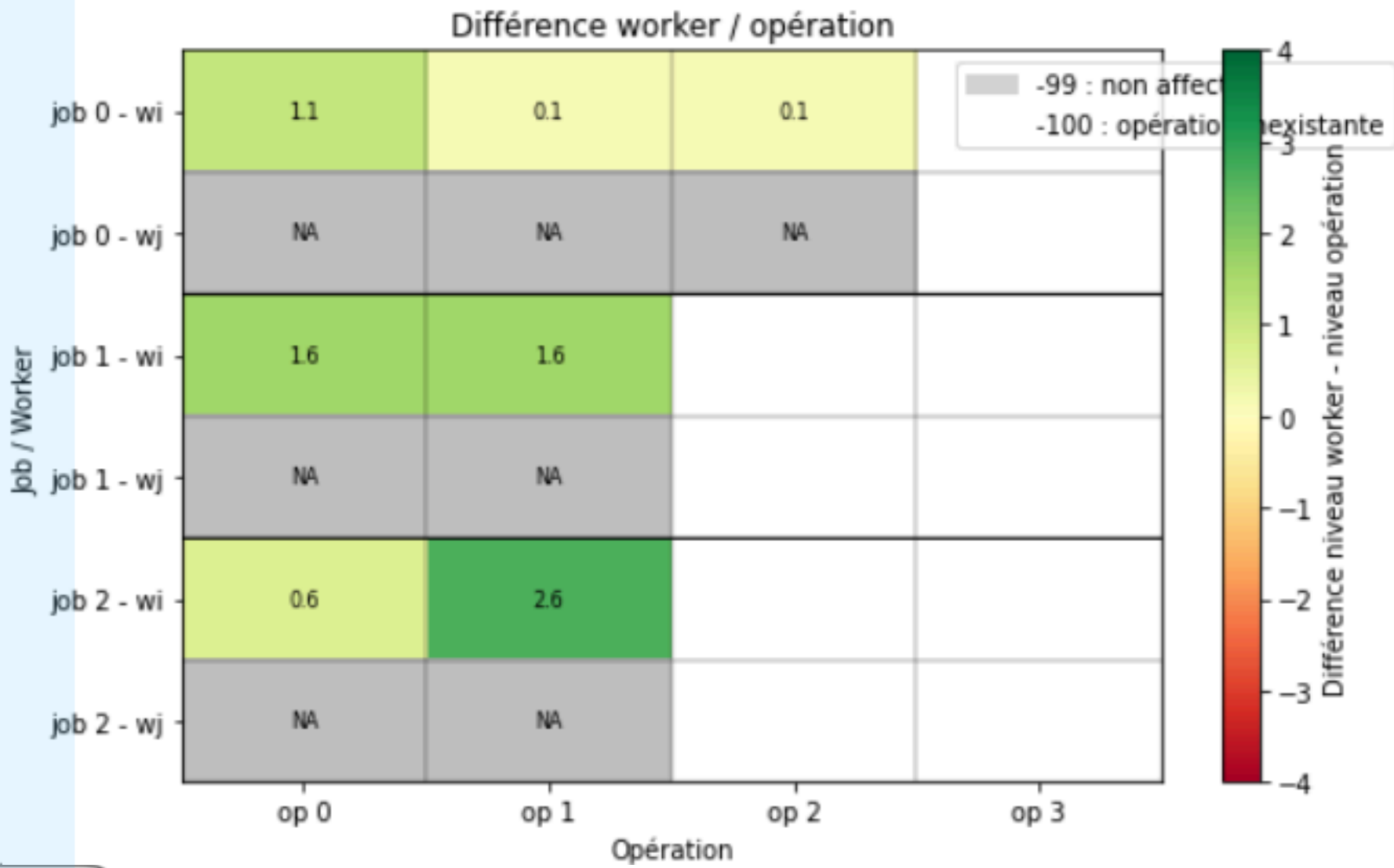
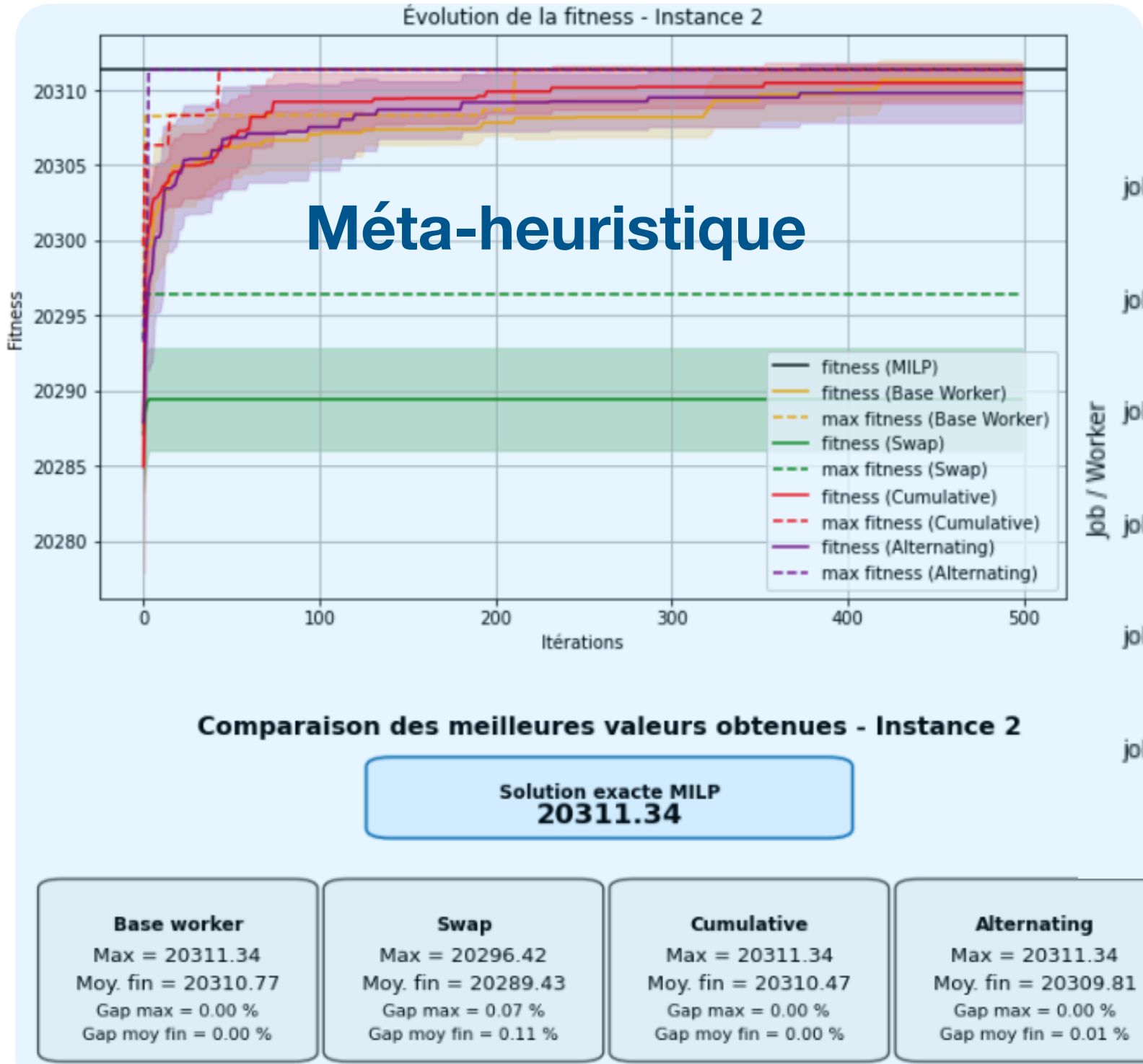
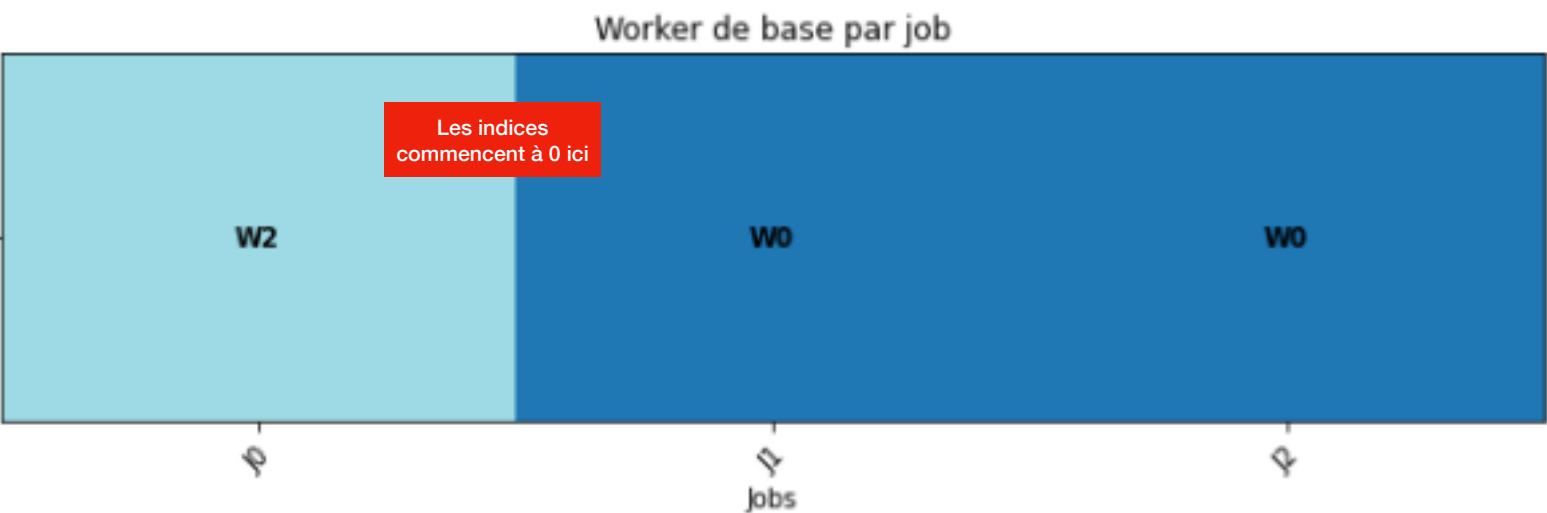
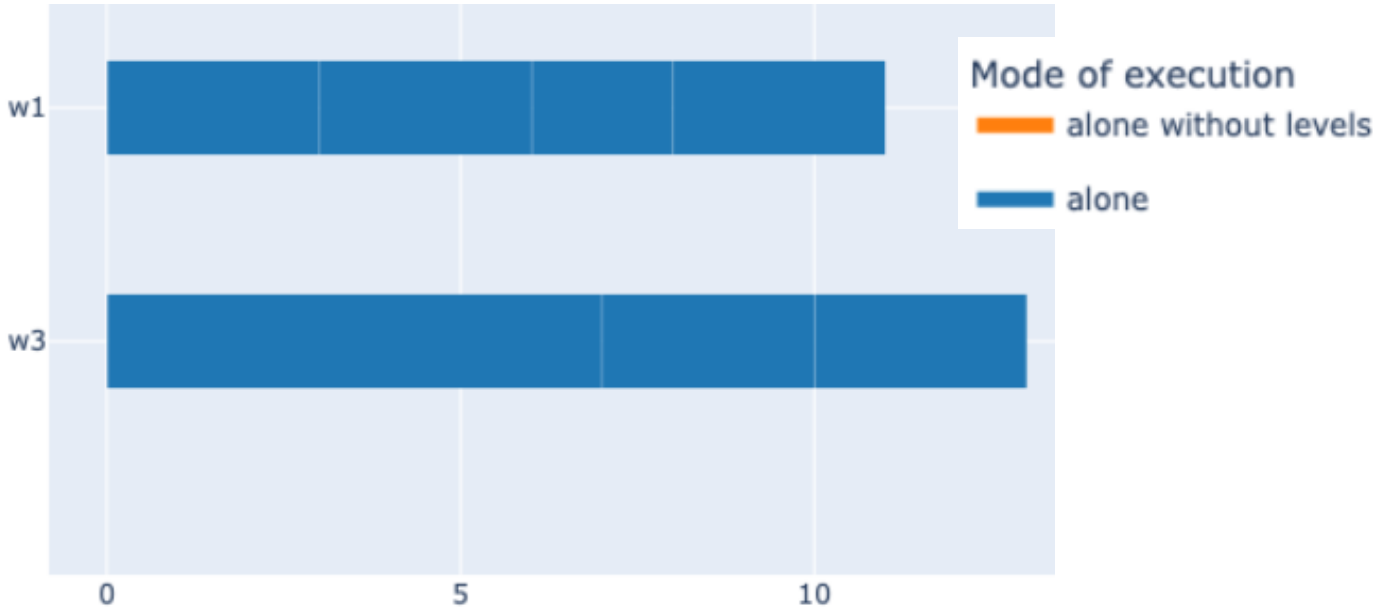
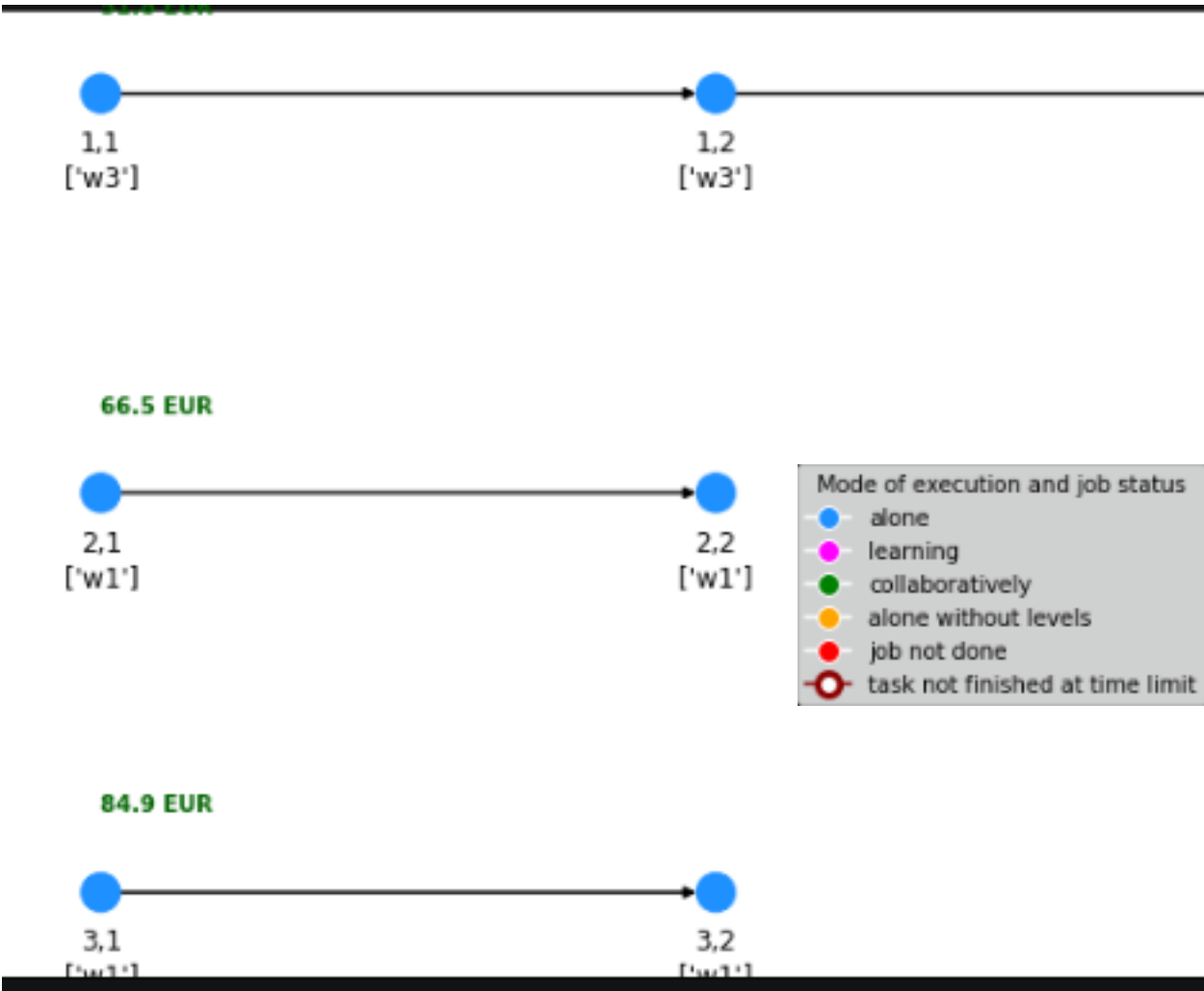
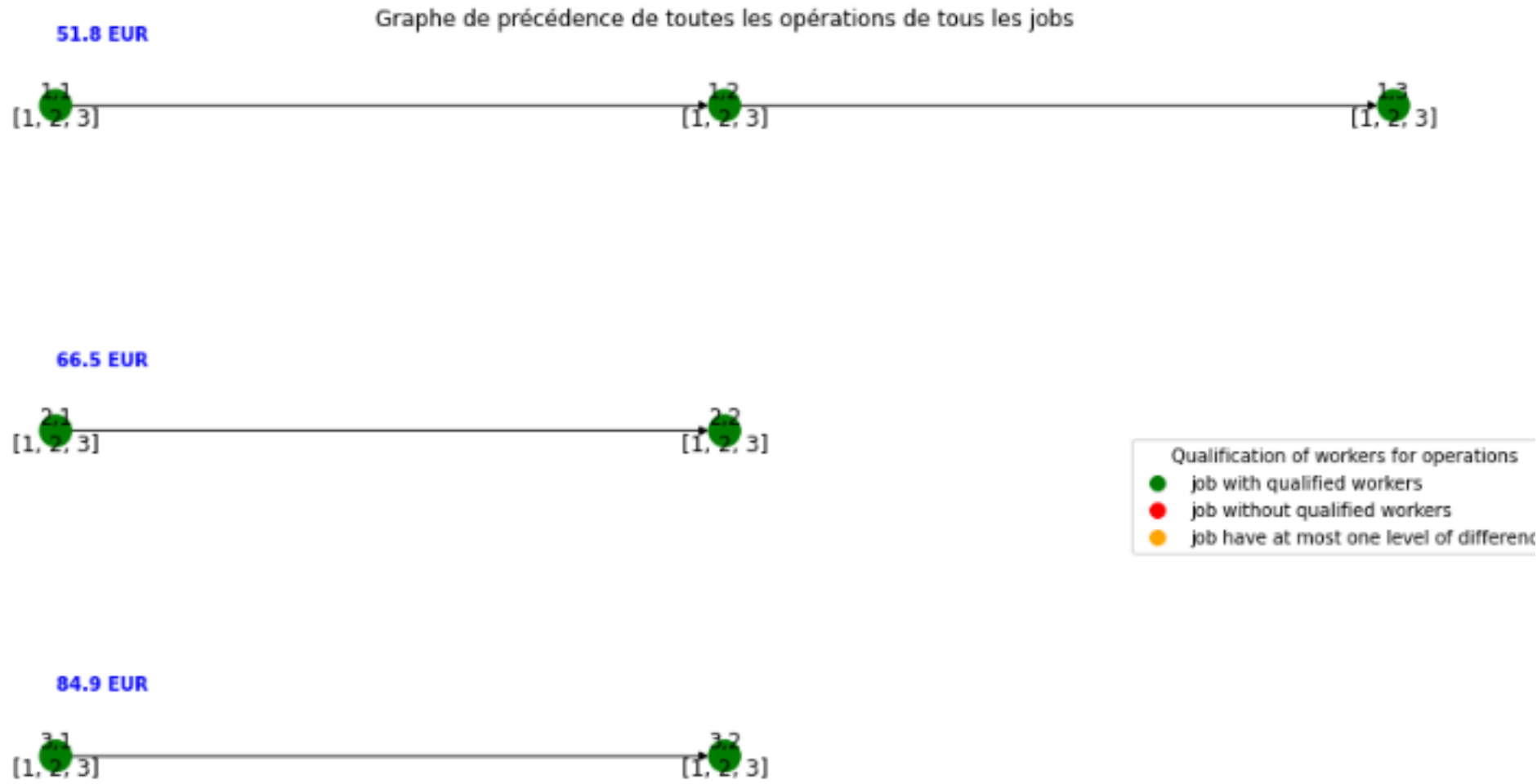
Alternating
Max = 17674.27
Moy. fin = 17672.53
Gap max = 0.06 %
Gap moy fin = 0.07 %



MILP MODEL

instance : 2
profit : 203.24341837462248
skills : 0.0
cognitive_load : 0.0
Cmax : 13.0
20311.34183746225

Instance 2



Comparaison des meilleures valeurs obtenues - Instance 2

Instance 2

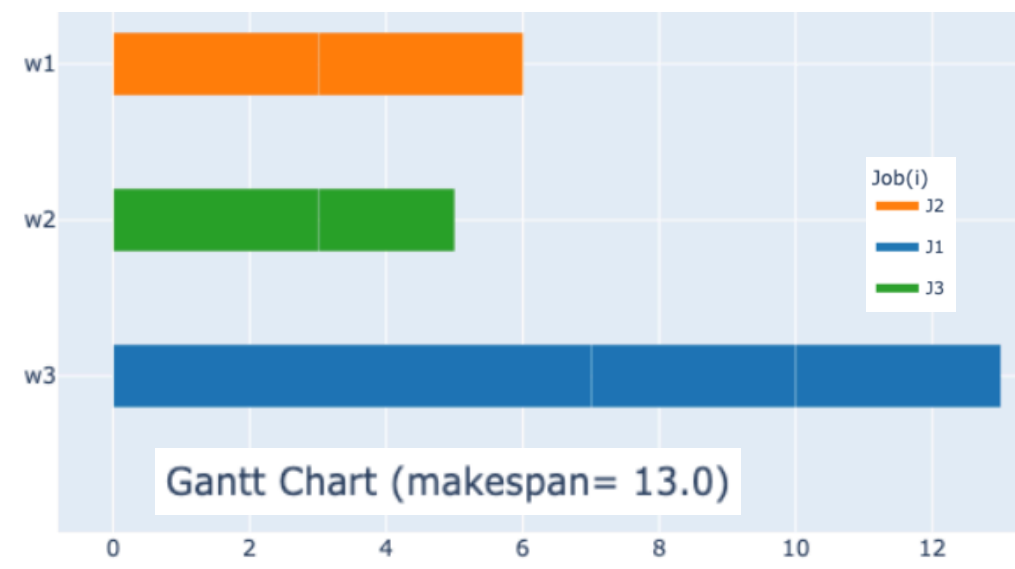
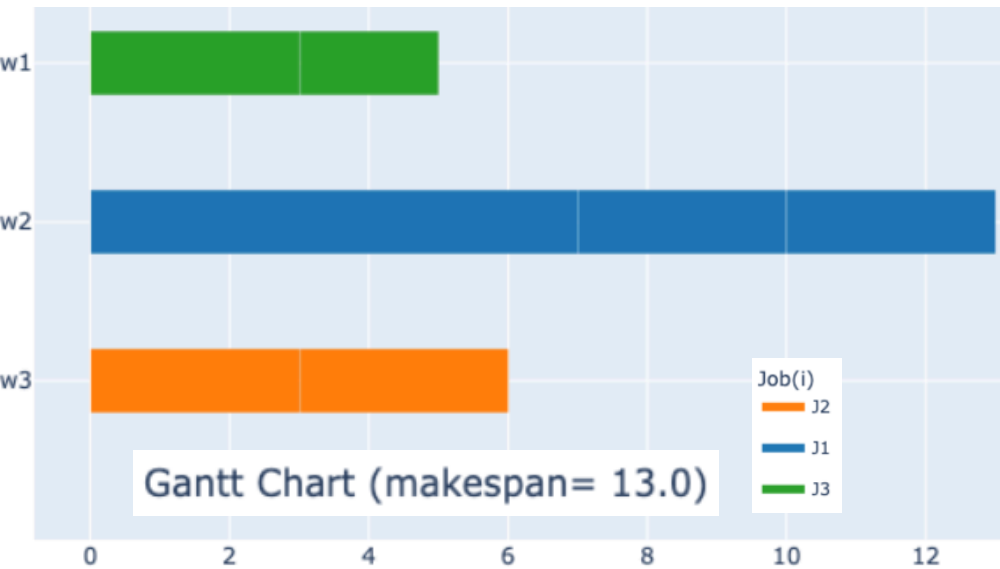
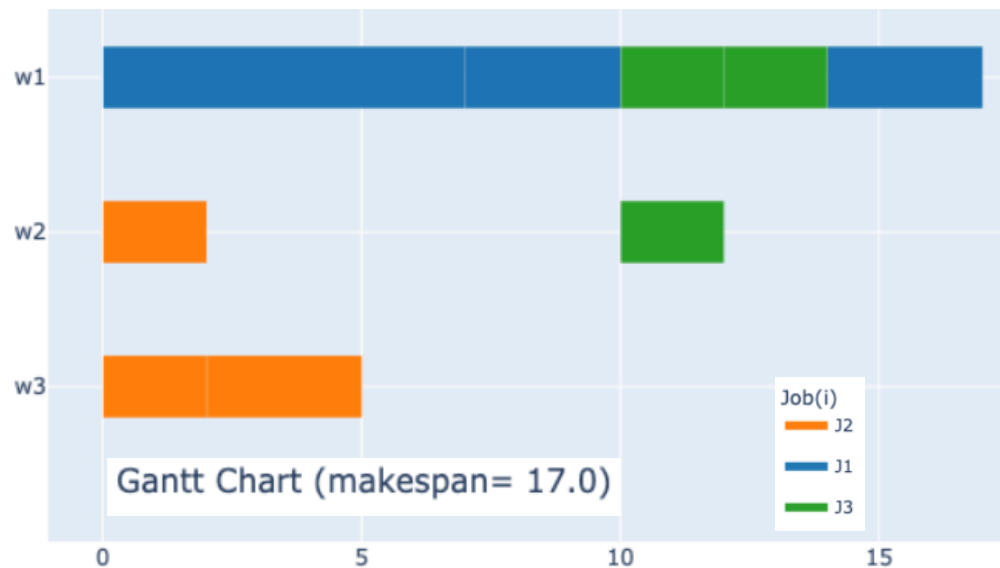
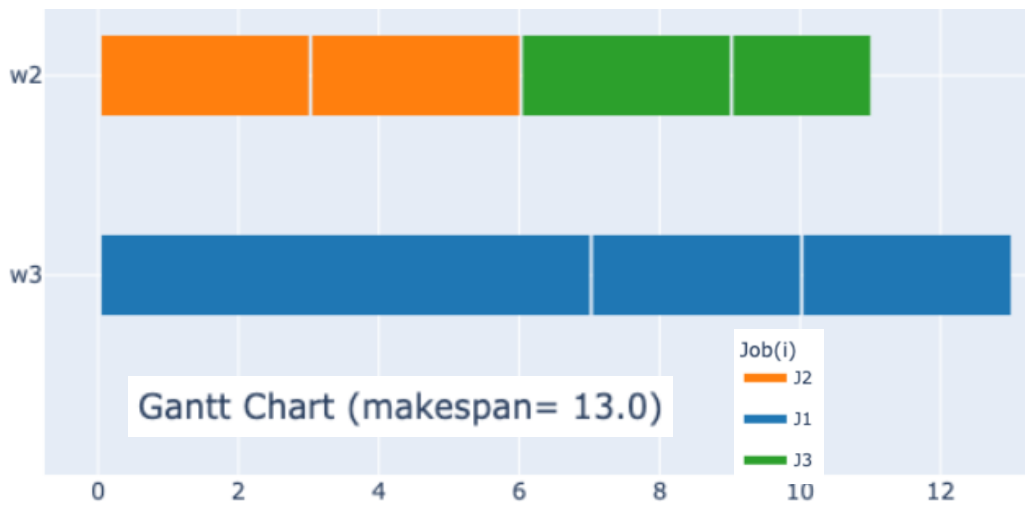
Solution exacte MILP
20311.34

Base worker
Max = 20311.34
Moy. fin = 20310.77
Gap max = 0.00 %
Gap moy fin = 0.00 %

Swap
Max = 20296.42
Moy. fin = 20289.43
Gap max = 0.07 %
Gap moy fin = 0.11 %

Cumulative
Max = 20311.34
Moy. fin = 20310.47
Gap max = 0.00 %
Gap moy fin = 0.00 %

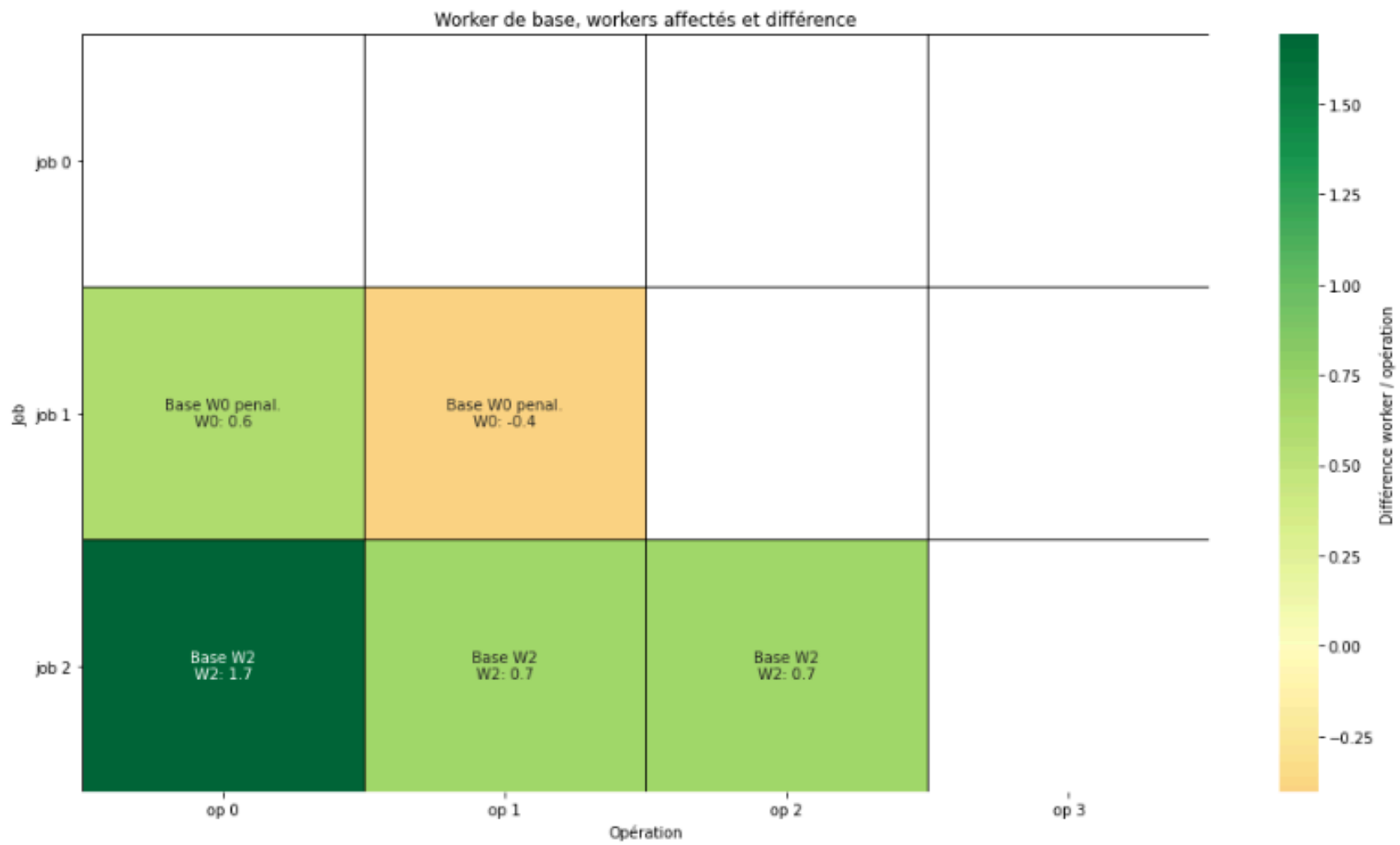
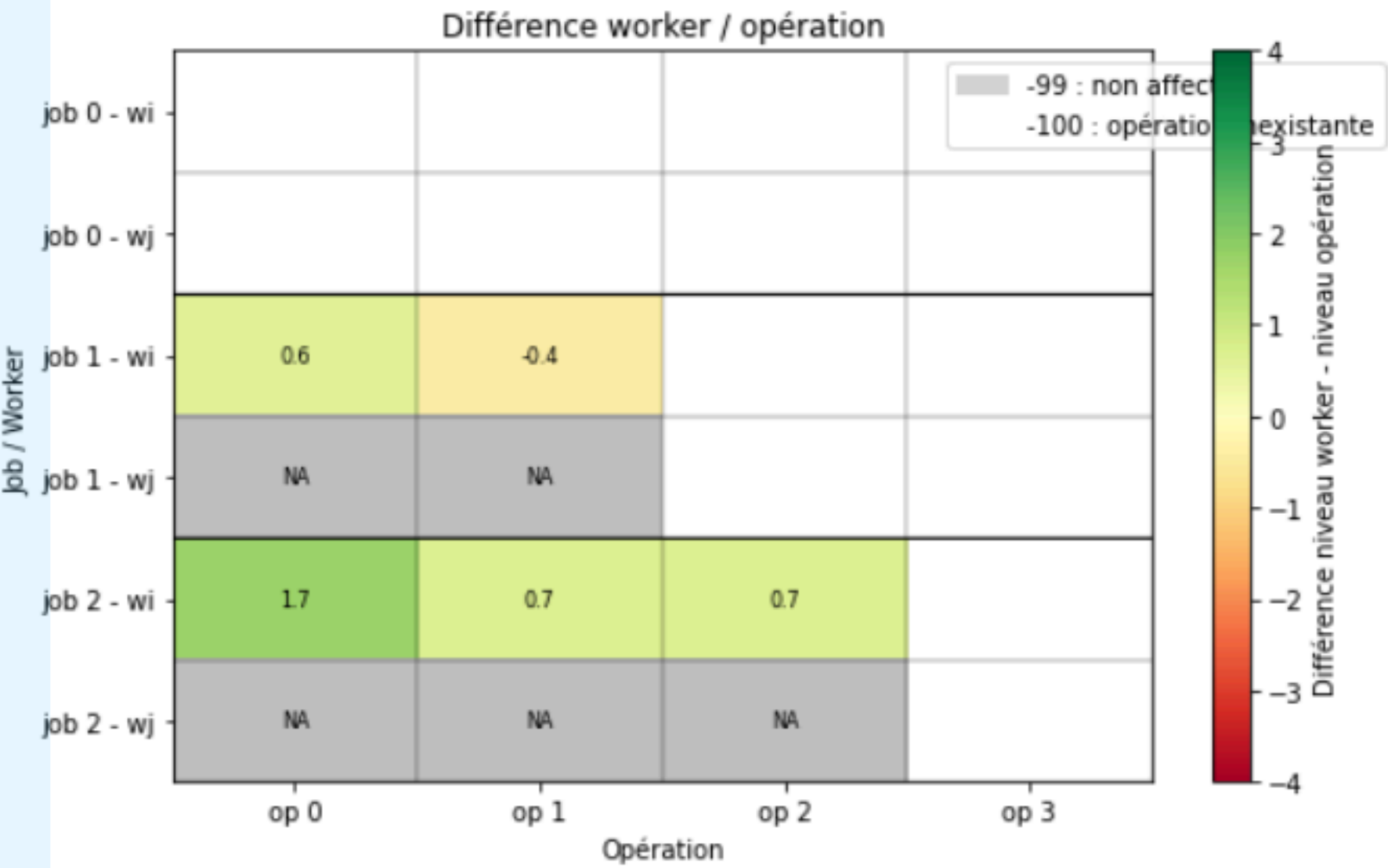
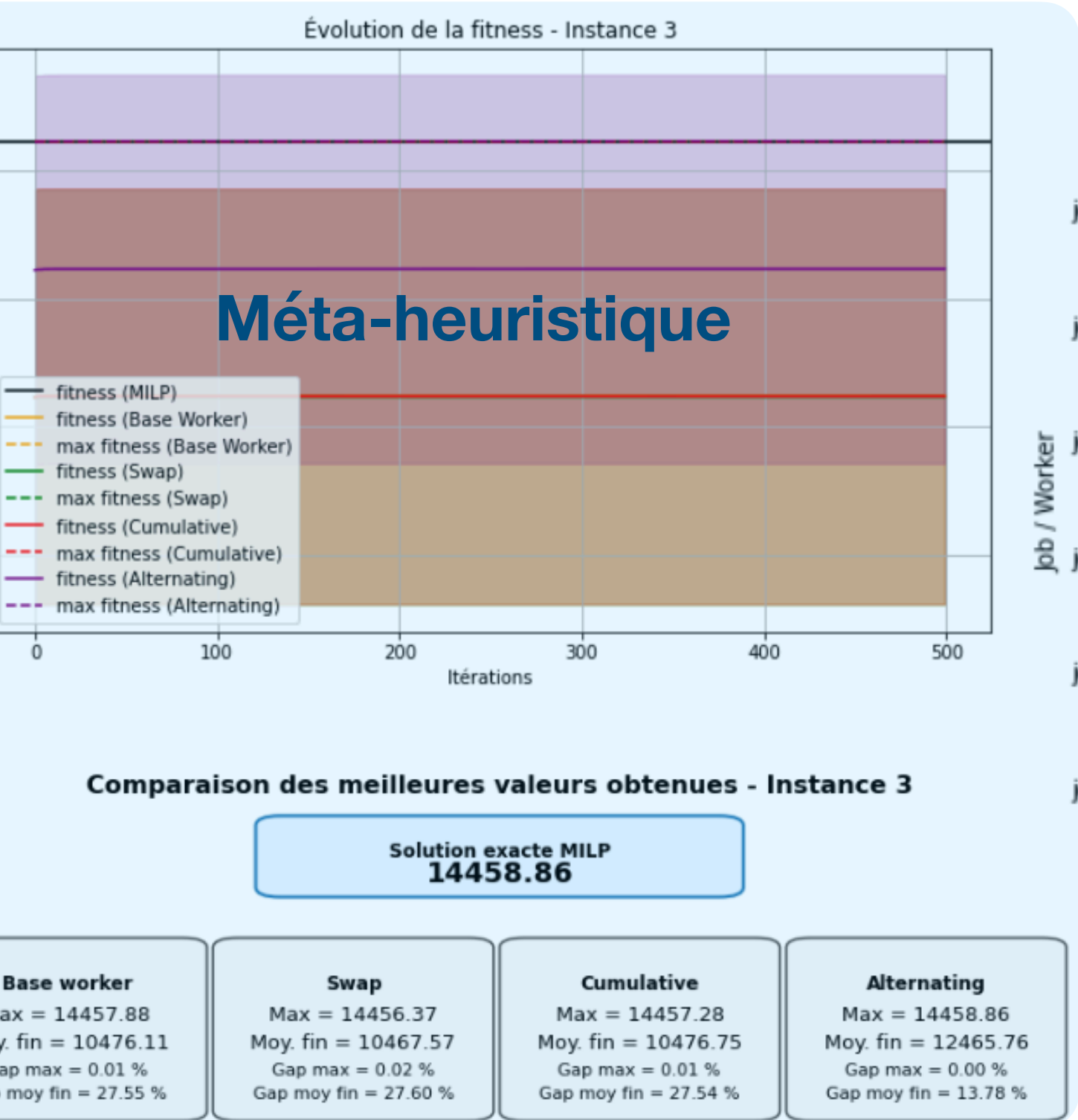
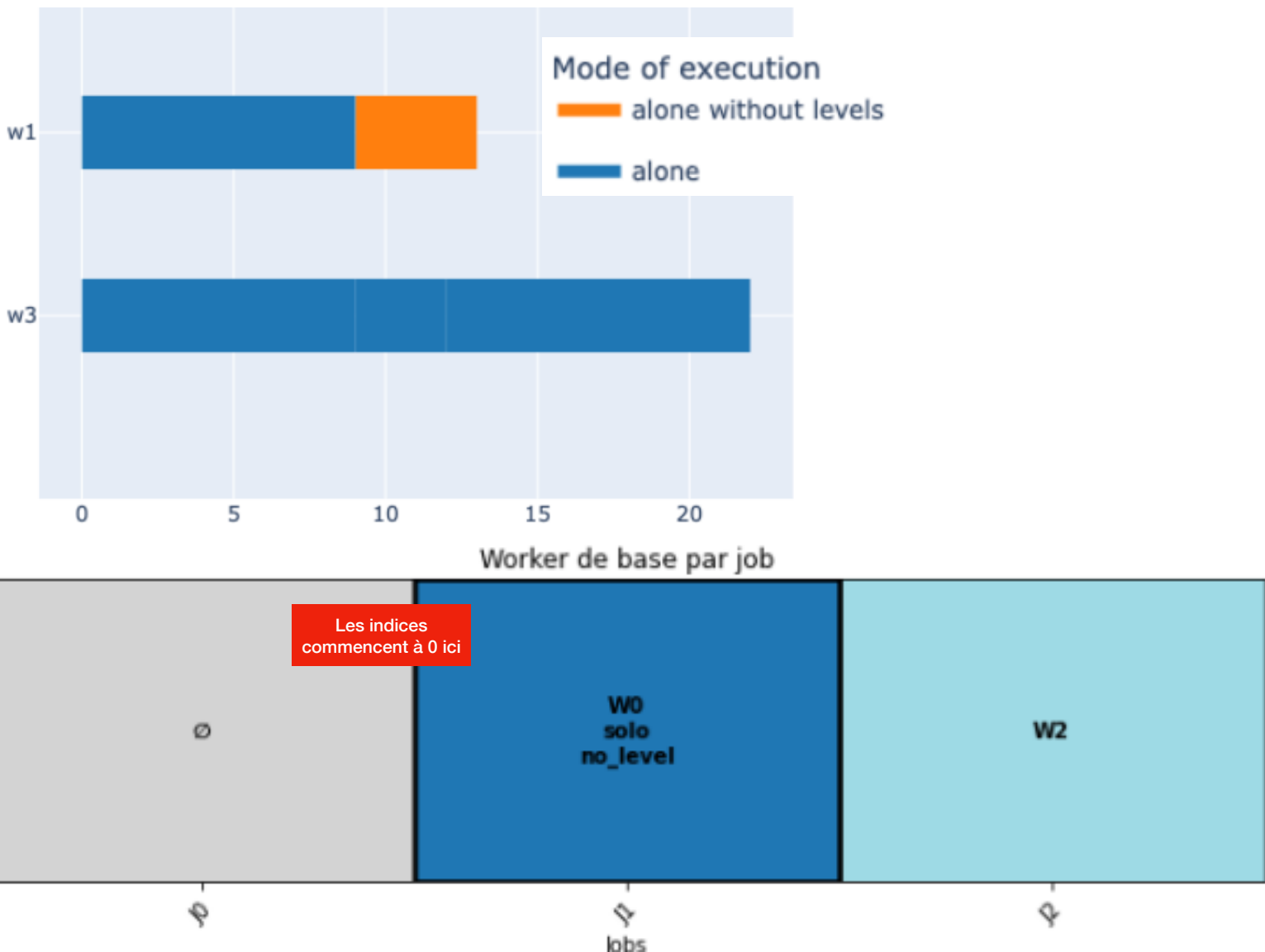
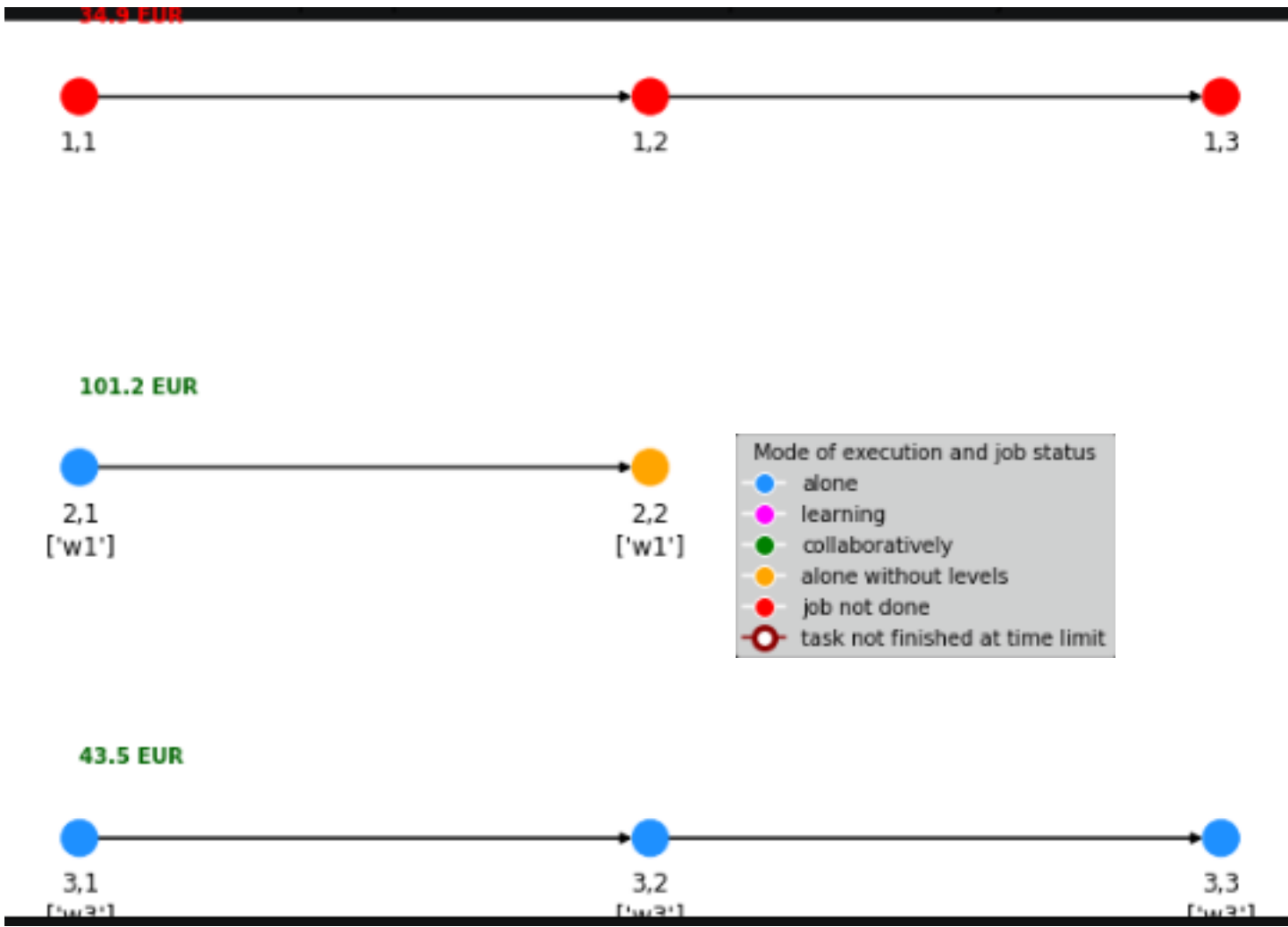
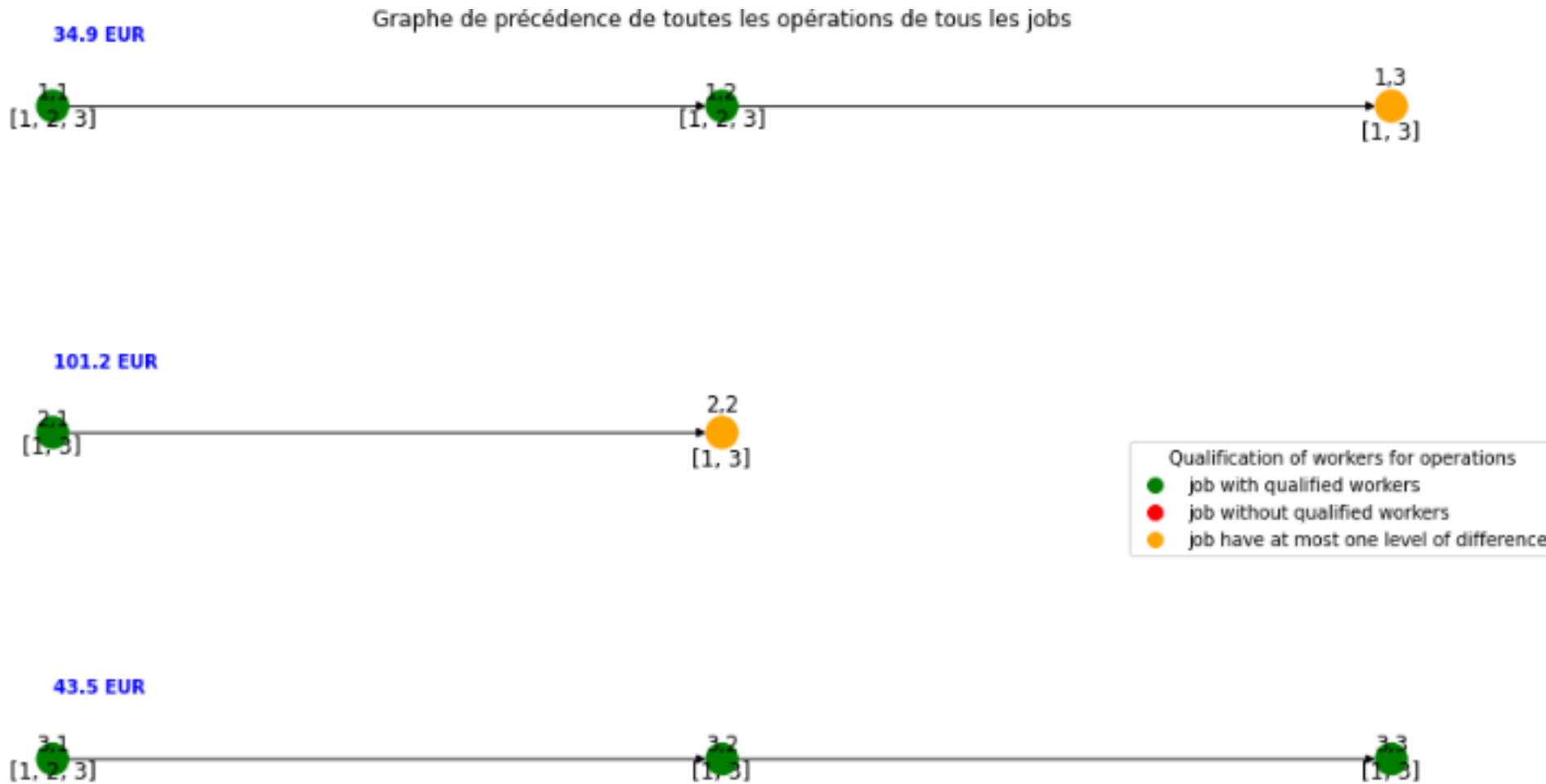
Alternating
Max = 20311.34
Moy. fin = 20309.81
Gap max = 0.00 %
Gap moy fin = 0.01 %



MILP MODEL

instance : 3
profit : 144.76806776826345
skills : 0.4053544369325994
cognitive_load : 0.0
Cmax : 22.0
14458.86032119567

Instance 3



Comparaison des meilleures valeurs obtenues - Instance 3

Solution exacte MILP
14458.86

Base worker

Max = 14457.88
Moy. fin = 10476.11
Gap max = 0.01 %
Gap moy fin = 27.55 %

Swap

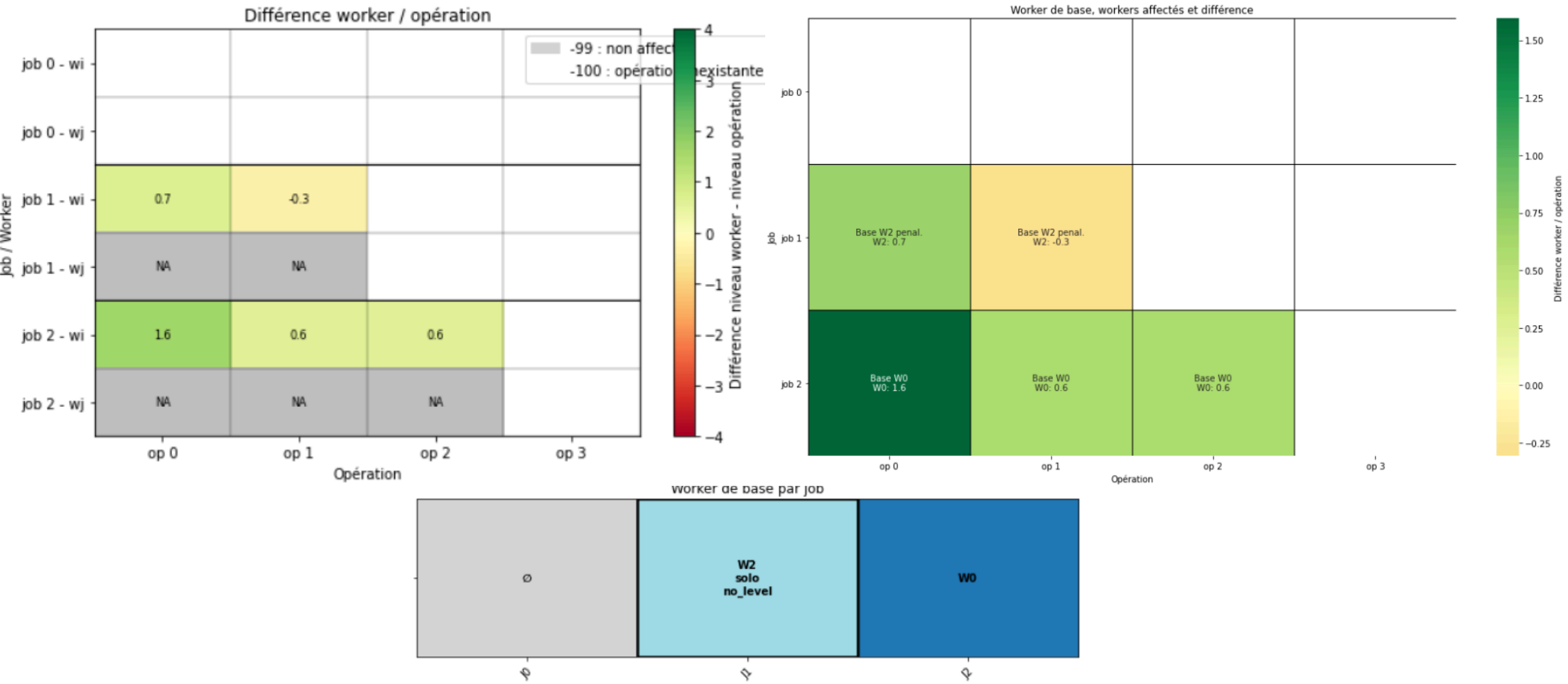
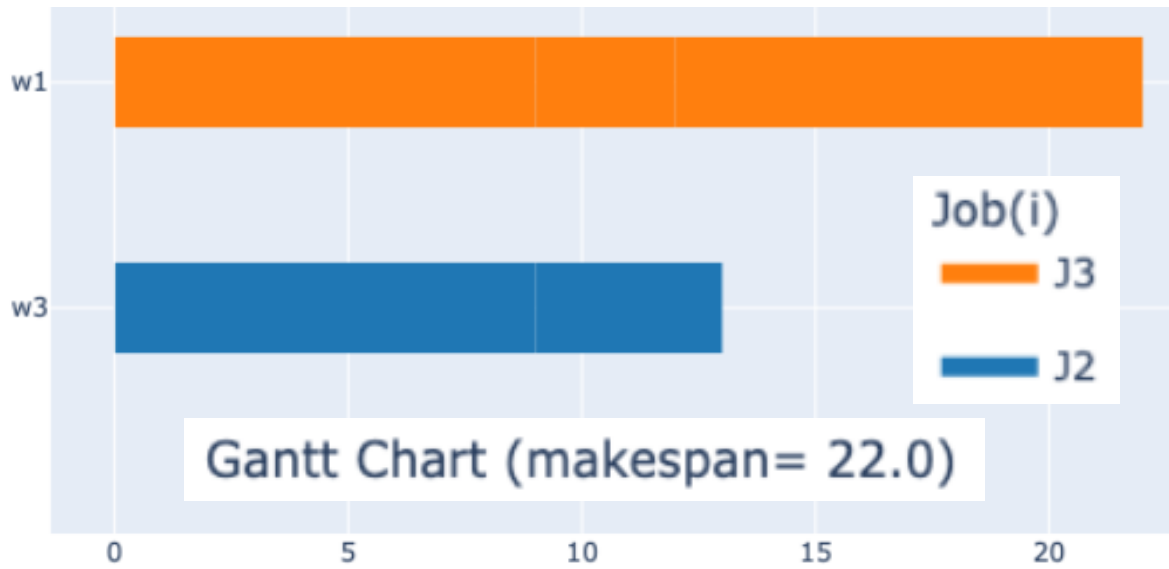
Max = 14456.37
Moy. fin = 10467.57
Gap max = 0.02 %
Gap moy fin = 27.60 %

Cumulative

Max = 14457.28
Moy. fin = 10476.75
Gap max = 0.01 %
Gap moy fin = 27.54 %

Alternating

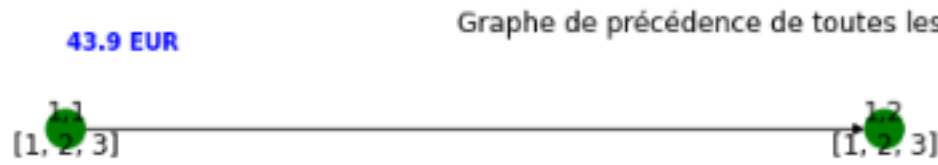
Max = 14458.86
Moy. fin = 12465.76
Gap max = 0.00 %
Gap moy fin = 13.78 %



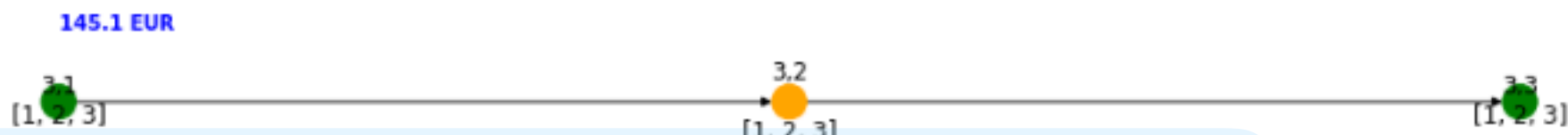
MILP MODEL

instance : 4
profit : 189.02667783439696
skills : 0.4732967096895724
cognitive_load : 0.0
Cmax : 32.0
18875.400750536595

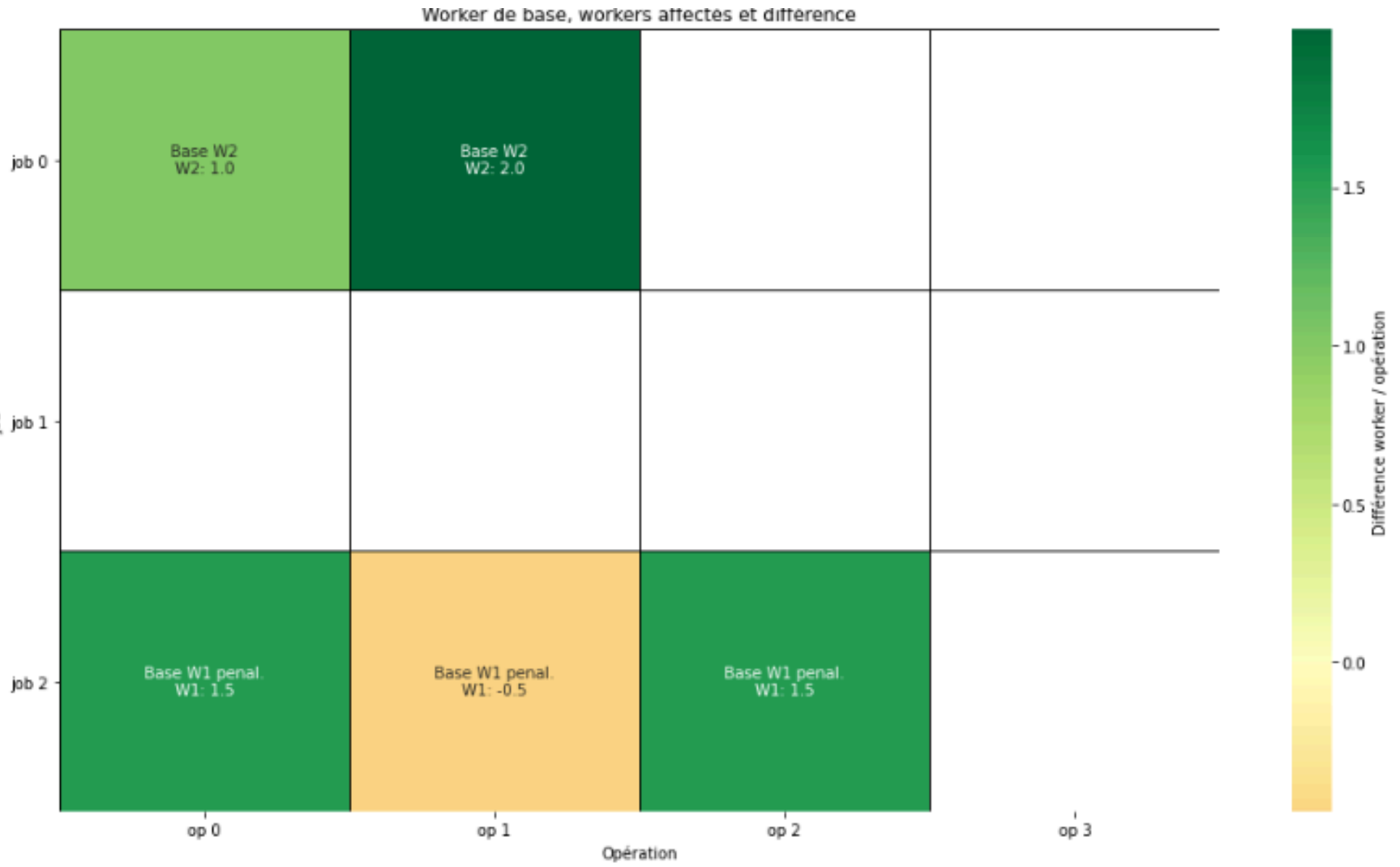
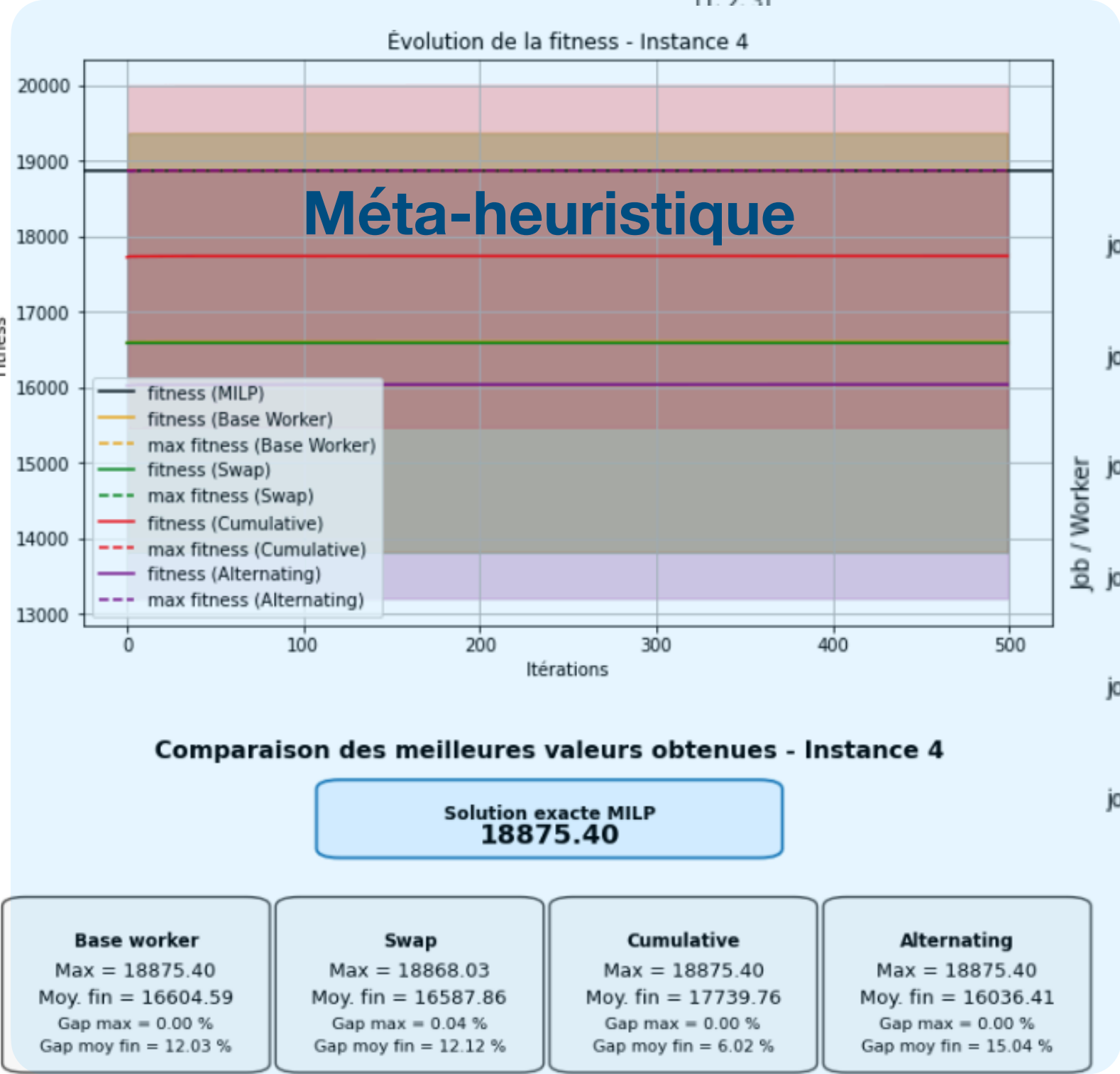
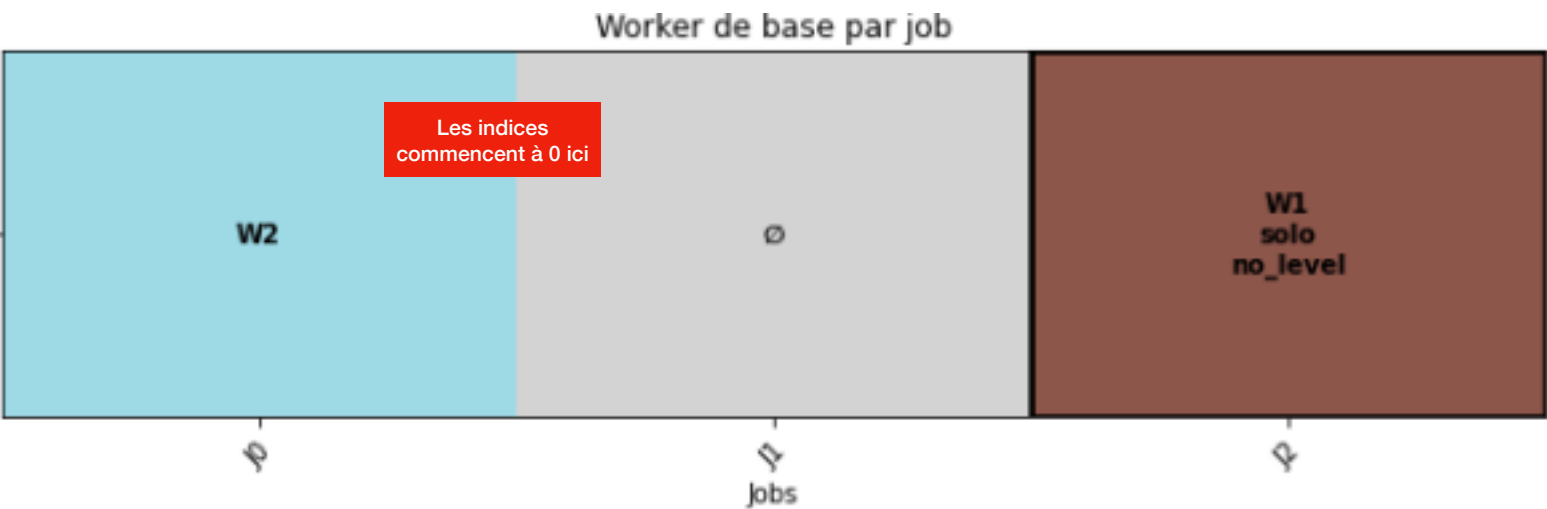
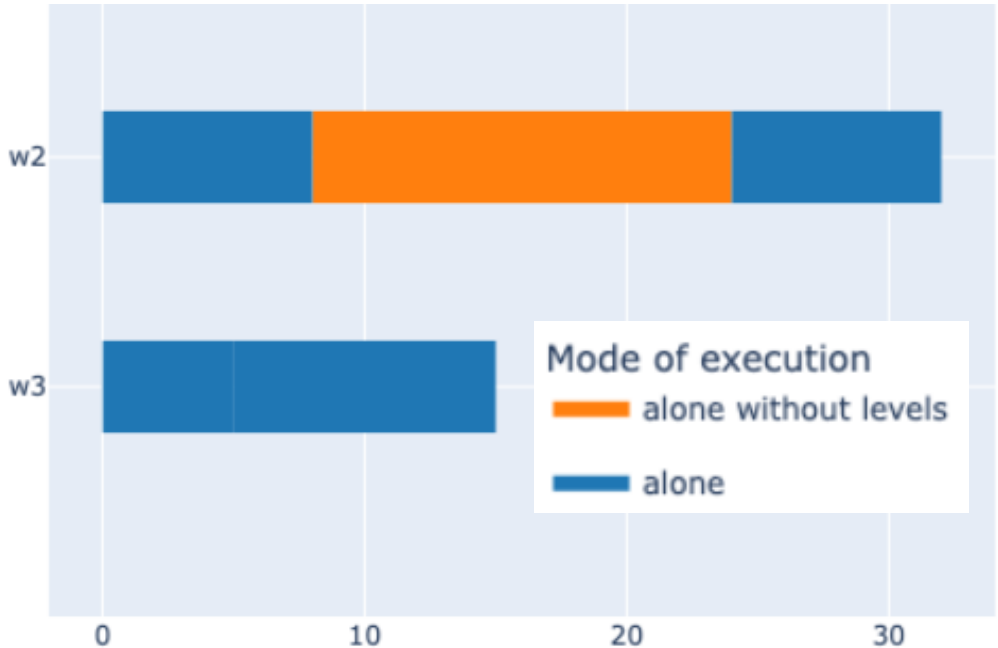
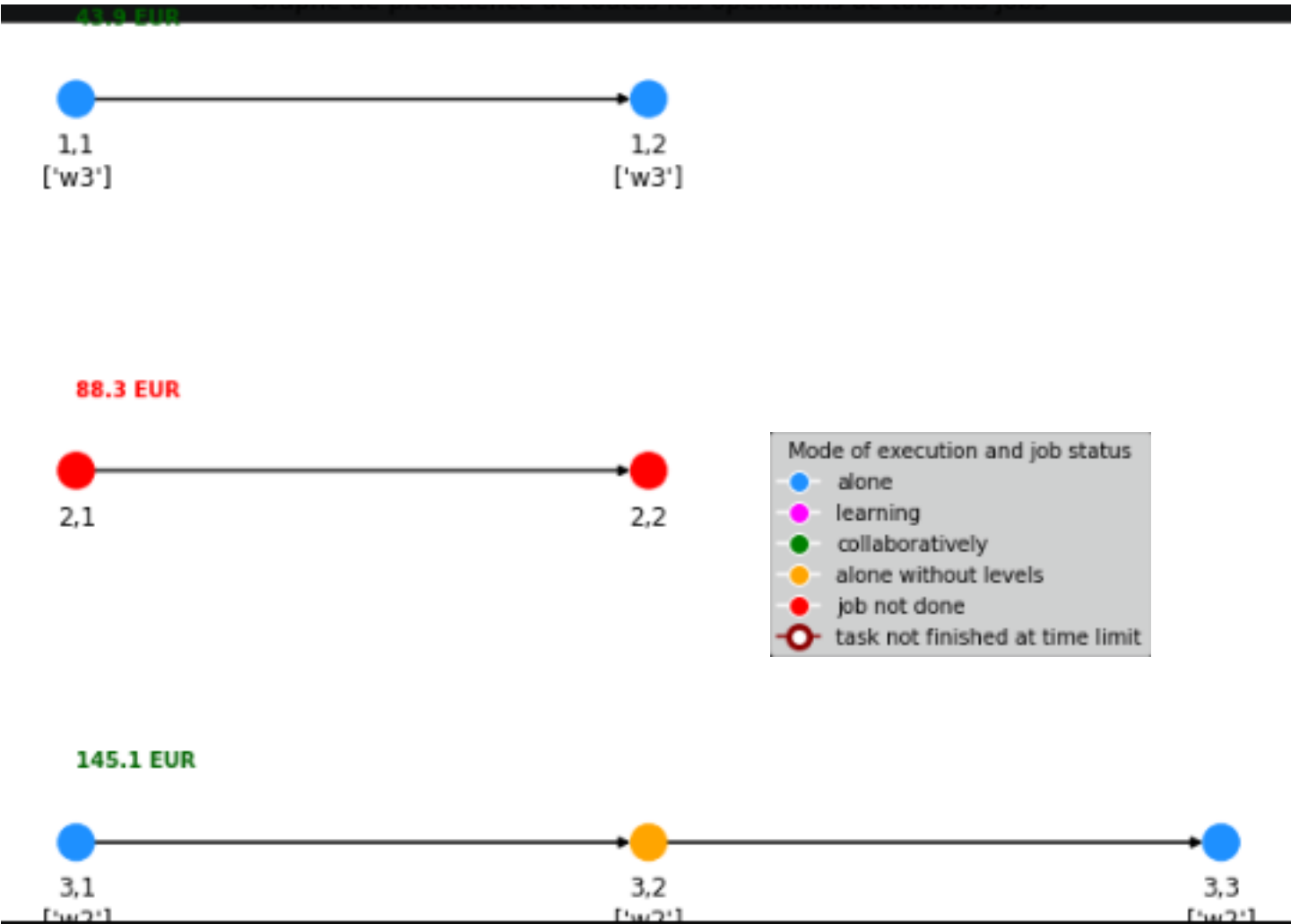
Instance 4



Graphe de précedence de toutes les opérations de tous les jobs



Qualification of workers for operations
● job with qualified workers
● job without qualified workers
● job have at most one level of difference



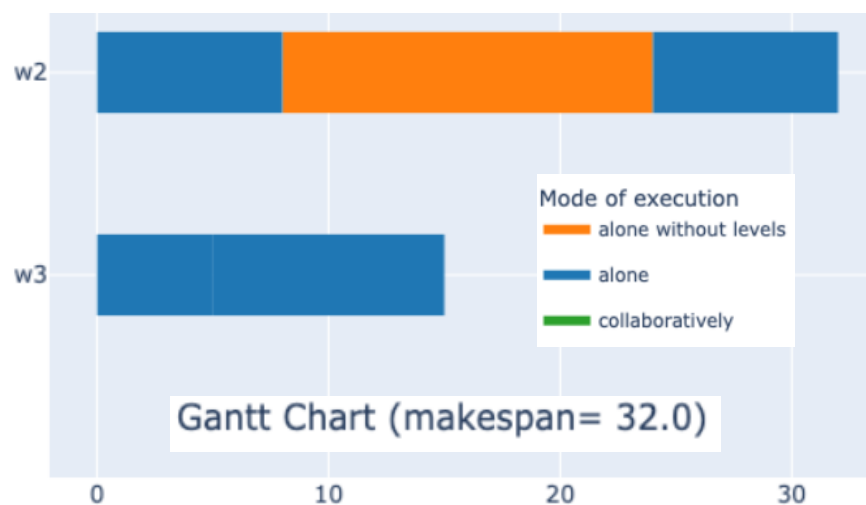
Comparaison des meilleures valeurs obtenues - Instance 4

Instance 4

Solution exacte MILP
18875.40

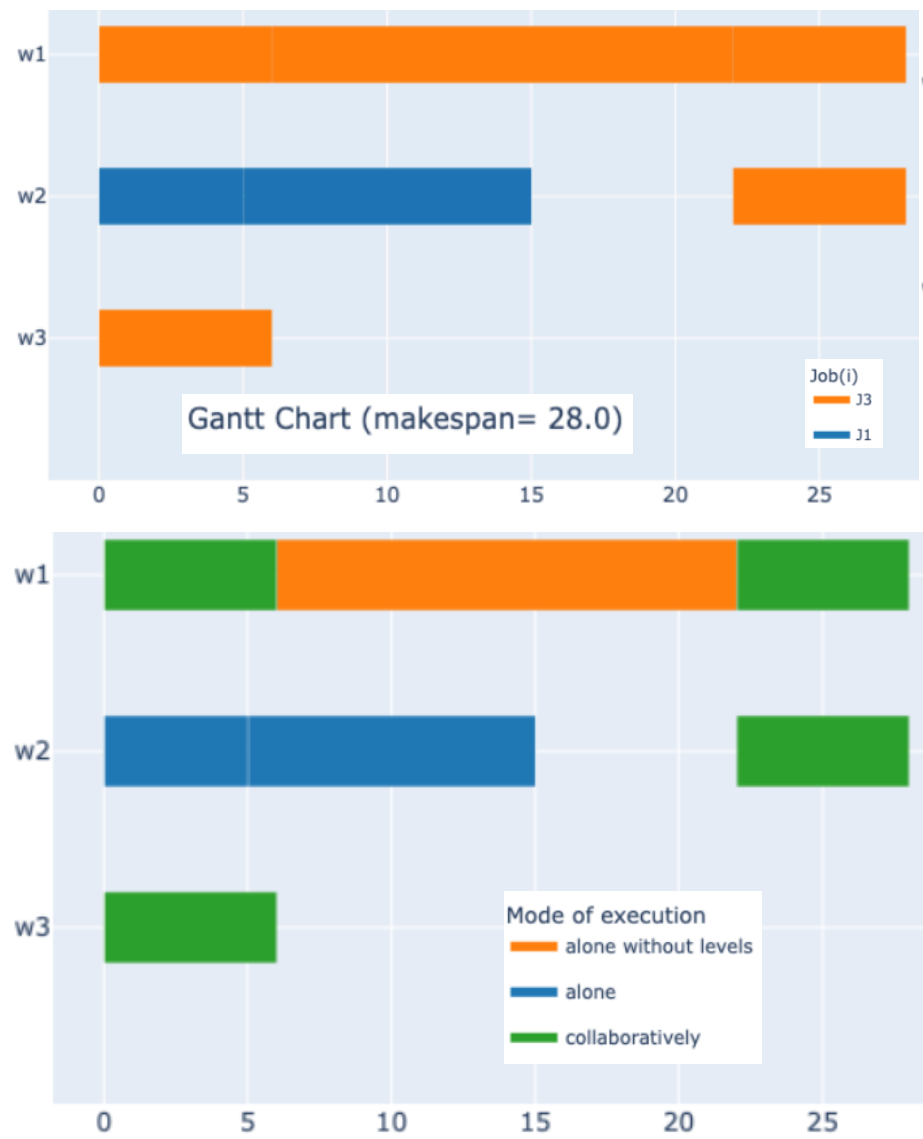
Base worker

Max = 18875.40
Moy. fin = 16604.59
Gap max = 0.00 %
Gap moy fin = 12.03 %



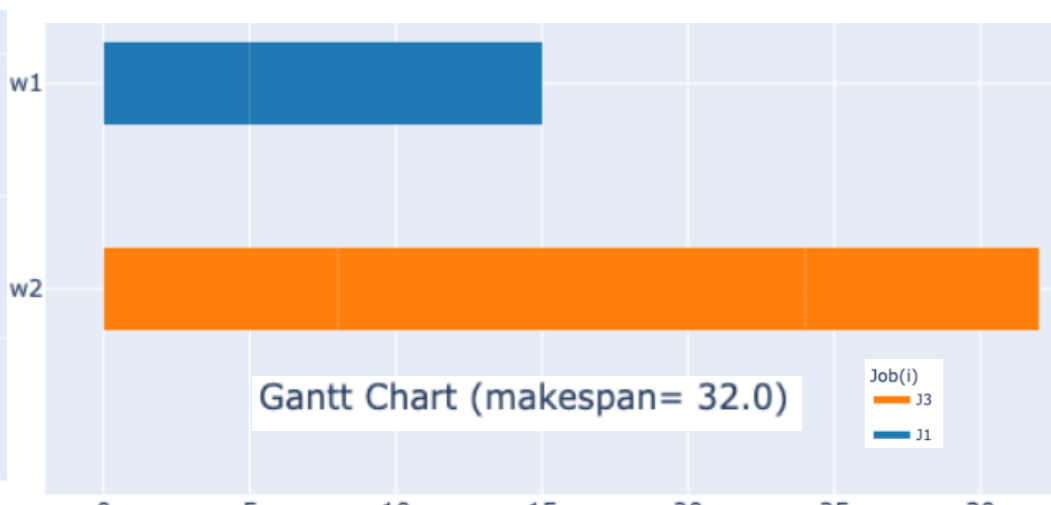
Swap

Max = 18868.03
Moy. fin = 16587.86
Gap max = 0.04 %
Gap moy fin = 12.12 %



Cumulative

Max = 18875.40
Moy. fin = 17739.76
Gap max = 0.00 %
Gap moy fin = 6.02 %



Alternating

Max = 18875.40
Moy. fin = 16036.41
Gap max = 0.00 %
Gap moy fin = 15.04 %

